# Quadrotor Optimal Control

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## Contents

**1 Introduction** **2**

**2 System Description and Dynamics** **3**

2.1 Frames and Conventions . . . 3
2.2 States, Inputs, and Parameters . . . 3
2.3 Kinematics . . . 4
2.4 Rigid–Body Dynamics . . . 4
2.5 Actuator Model and Allocation . . . 5
2.6 Hover Equilibrium . . . 6
2.7 Compact Nonlinear Model Summary . . . 6
2.8 Modeling Assumptions and Possible Extensions . . . 6

**3 Minimum Principle Formulation** **7**

3.1 Dynamics and Cost . . . 7
3.2 Hamiltonian, Stationarity, and Boundary Conditions . . . 7
3.3 Costate Equations . . . 8
3.4 Closed-Form Control Components . . . 9
3.5 Boundary and Transversality Conditions . . . 9

**4 Linearization** **9**

4.1 Hovering Equilibrium . . . 9
4.2 Controllability . . . 11

**5 Minimum Principle on the Linearized Model** **11**

5.1 Finite-horizon quadratic problem . . . 11
5.2 Actuator limits . . . 12
5.3 Minimum-time variant (bang–bang) . . . 13

5.4 Infinite-horizon case: boundary condition and adjusted TPBVP . . . 13

**6 Linear–Quadratic Regulator (LQR) 13**

6.1 Finite-horizon LQR . . . 14

6.2 Infinite-horizon LQR . . . 14

**7 Model Predictive Control 14**

7.1 Linear MPC . . . 15

7.2 Nonlinear MPC . . . 16

**8 Simulations and Results 17**

8.1 System Parameters . . . 17

8.2 The dynamic model . . . 18

8.3 Linearized Model at Hover . . . 18

8.4 LQR . . . 19

8.5 Linear MPC . . . 21

8.6 Nonlinear MPC . . . 23

8.7 Sensitivity to Horizon and Actuator Limits . . . 25

8.8 Minimum Principle TPBVP . . . 28

8.9 Summary and Comparison . . . 30

**9 Conclusion 31**

# 1 Introduction

Quadrotors are an instructive testbed for comparing linear and nonlinear optimal control since the rigid–body dynamics are well understood while the actuation map from rotor speeds to thrust–torque is quadratic and subject to hard bounds. We study position–yaw step tracking near hover with four controllers that share the same dynamics and quadratic cost $(Q, R)$—an infinite–horizon LQR designed on the hover linearization, a linear MPC in deviation form using a ZOH–discretized model with a DARE terminal weight and input boxes induced by rotor–speed limits, a nonlinear MPC on the full model solved by direct multiple shooting with squared motor speeds as the stage decision, and a nonlinear optimal control law from Pontryagin's Minimum Principle solved by direct multiple shooting with an interior–point method. All controllers enforce identical hard rotor–speed bounds, consistently mapped into the thrust–torque input space via the same allocation.

A standard 12–state rigid–body formulation in a North–East–Down frame with ZYX Euler angles is adopted. The control inputs are total thrust and body torques, realized through a quadratic mapping from squared rotor speeds subject to hard RPM limits. Linearization about hover is performed to obtain $(A, B)$, and a single quadratic performance index with diagonal weights $(Q, R)$ is employed for both methods. The tracking task is specified as a single step in $[p_x, p_y, p_z, \psi]$ with all other references set to zero; roll and pitch are required to return to zero.

## 2 System Description and Dynamics

This section defines a continuous–time nonlinear model of a quadrotor UAV. ZYX (yaw–pitch–roll) Euler angles are used.

### 2.1 Frames and Conventions

- **Inertial frame (NED)** $\{\mathcal{N}\}$: right–handed with $x_N$ North, $y_E$ East, $z_D$ Down. Gravity is $g > 0$ acting *along* $+z_D$, i.e., $\boldsymbol{g} = g\,[0,\ 0,\ 1]^\top$.
- **Body frame** $\{\mathcal{B}\}$: right–handed with $x$ forward, $y$ to the right, $z$ downward.
- **Euler angles (ZYX)**: roll $\phi$ about $x$, pitch $\theta$ about $y$, yaw $\psi$ about $z$. The body to inertial transformation is written in map notation as

$$ {}^{\mathcal{N}}R_{\mathcal{B}} : \mathbb{R}^3 \to \mathrm{SO}(3), \qquad (\phi,\theta,\psi) \ \mapsto\ R(\phi,\theta,\psi) := R_z(\psi)\,R_y(\theta)\,R_x(\phi). \tag{2.1} $$

We use

$$ R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix},\ R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix},\ R_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{2.2} $$

*Remark.* This choice matches common aerospace texts; if a robotics/ROS ENU convention is preferred ($x$ forward, $y$ left, $z$ up), one replaces $\boldsymbol{g} = [0, 0, -g]^\top$ and flips the thrust sign in Eq. (2.8).

### 2.2 States, Inputs, and Parameters

We use a 12–state representation

$$ \boldsymbol{x} = \begin{bmatrix} \boldsymbol{p} \\ \boldsymbol{v} \\ \boldsymbol{\eta} \\ \boldsymbol{\omega} \end{bmatrix} \in \mathbb{R}^{12}, \qquad \boldsymbol{p} = \begin{bmatrix} x & y & z \end{bmatrix}^\top, \quad \boldsymbol{v} = \begin{bmatrix} \dot{x} & \dot{y} & \dot{z} \end{bmatrix}^\top, \tag{2.3} $$

$$ \boldsymbol{\eta} = \begin{bmatrix} \phi & \theta & \psi \end{bmatrix}^\top, \quad \boldsymbol{\omega} = \begin{bmatrix} p & q & r \end{bmatrix}^\top \tag{2.4} $$

where $\boldsymbol{p}$ is the position of the body in $\mathcal{N}$, $\boldsymbol{v}$ is the translational velocity of the body expressed in $\mathcal{N}$, $\boldsymbol{\eta}$ is the angular position of the body in $\mathcal{N}$, and $\boldsymbol{\omega}$ is the body angular rates in $\mathcal{B}$. The control input is the body wrench

$$ \boldsymbol{u} = \begin{bmatrix} T & \tau_x & \tau_y & \tau_z \end{bmatrix}^\top, \tag{2.5} $$

where $T \geq 0$ is the *thrust magnitude*. In the NED convention thrust acts *opposite* the body $+z$ axis, so the force in $\{\mathcal{N}\}$ is $-T\,\boldsymbol{b}_3$ with $\boldsymbol{b}_3$ the third column of $R$ (Eq. (2.1)). Model parameters: mass $m > 0$, inertia $J = \mathrm{diag}(J_x, J_y, J_z) \succ 0$, *non-symmetric* axis lever arms $L_x > 0$ (forward/back component) and $L_y > 0$

(right/left component), thrust coefficient $k_f > 0$, yaw torque coefficient $c_\tau > 0$. Fig. 1 shows the quadrotor rigid body and the used frame system.

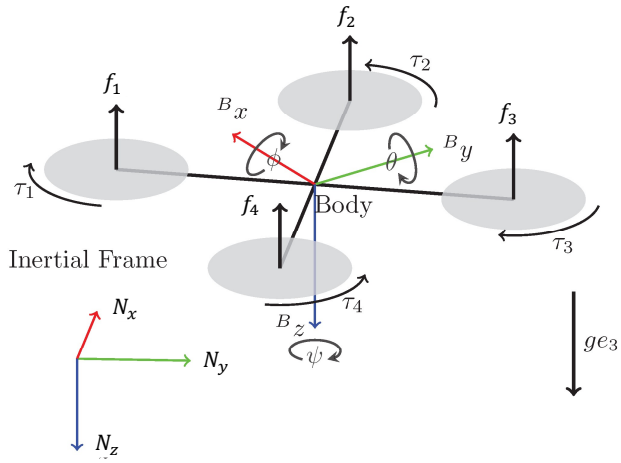


**Figure 1:** Quadrotor rigid body and used frame system.

## 2.3 Kinematics

Translational kinematics are

$$\dot{\boldsymbol{p}} = \boldsymbol{v}. \tag{2.6}$$

Euler angle rates relate to body rates through

$$\dot{\boldsymbol{\eta}} = W(\phi, \theta)\, \boldsymbol{\omega}, \qquad W(\phi, \theta) = \begin{bmatrix} 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi/\cos\theta & \cos\phi/\cos\theta \end{bmatrix}, \tag{2.7}$$

which is well-defined for $\cos\theta \neq 0$.

## 2.4 Rigid–Body Dynamics

**Translational dynamics.** Let $\boldsymbol{e}_3 = [0,\ 0,\ 1]^\top$ denote the *down* axis of NED and let $\boldsymbol{b}_3$ be the third column of $R$ (body $z$ in $\{\mathcal{N}\}$). Then

$$m\,\dot{\boldsymbol{v}} = m\,g\,\boldsymbol{e}_3 \;-\; T\,\boldsymbol{b}_3(\eta) \quad \Longrightarrow \quad \dot{\boldsymbol{v}} = g\,\boldsymbol{e}_3 - \frac{T}{m}\,\boldsymbol{b}_3(\eta). \tag{2.8}$$

where

$$\boldsymbol{b}_3(\eta) = R(\eta)\,\boldsymbol{e}_3 \;=\; \begin{bmatrix} \sin\phi\sin\psi + \sin\theta\cos\phi\cos\psi \\ -\sin\phi\cos\psi + \sin\psi\sin\theta\cos\phi \\ \cos\phi\cos\theta \end{bmatrix}. \tag{2.9}$$

**Rotational dynamics.** With body rates $\boldsymbol{\omega} = [p,\, q,\, r]^\top$,

$$\dot{\boldsymbol{\omega}} = J^{-1}(\boldsymbol{\tau} - \boldsymbol{\omega} \times (J\,\boldsymbol{\omega}))\,. \tag{2.10}$$

Equations Eq. (2.6)–Eq. (2.10) define the nonlinear vector field used throughout.

## 2.5 Actuator Model and Allocation

**Numbering and geometry.** Motor order (fixed): **1=FR**, **2=RR**, **3=RL**, **4=FL**. Let $L_x > 0$ be the front–back motor spacing (pitch axis length) and $L_y > 0$ the right–left spacing (roll axis length). Rotor positions in $\{\mathcal{B}\}$ are

$$\boldsymbol{r}_1 = [+L_x/2, +L_y/2, 0]^\top, \quad \boldsymbol{r}_2 = [-L_x/2, +L_y/2, 0]^\top, \quad \boldsymbol{r}_3 = [-L_x/2, -L_y/2, 0]^\top, \quad \boldsymbol{r}_4 = [+L_x/2, -L_y/2, 0]^\top.$$

**Forces, spins, and wrench.** Each rotor produces thrust $f_i \geq 0$ along $-\boldsymbol{e}_3$ (opposite $+z_{\mathcal{B}}$). The thrust moment from $\boldsymbol{r}_i$ is $\boldsymbol{r}_i \times (-f_i \boldsymbol{e}_3)$, giving $\tau_x^{(i)} = -y_i f_i$, $\tau_y^{(i)} = +x_i f_i$. Yaw comes from prop drag: $\pm c_\tau f_i$ about $+z_{\mathcal{B}}$. Fig. 2 shows the rotor ordering and their respective spinning direction.

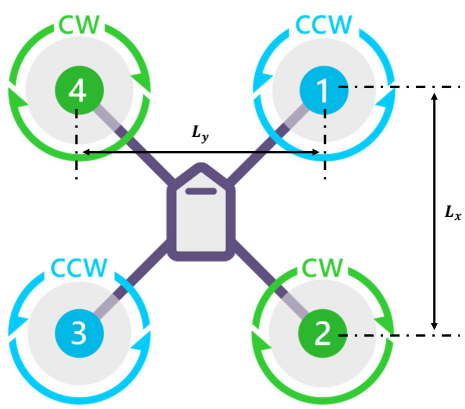


**Figure 2:** Rotors ordering and spinning direction.

Stacking $\boldsymbol{f} = [f_1\ f_2\ f_3\ f_4]^\top$, the mapping is

$$\begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\frac{L_y}{2} & -\frac{L_y}{2} & +\frac{L_y}{2} & +\frac{L_y}{2} \\ +\frac{L_x}{2} & -\frac{L_x}{2} & -\frac{L_x}{2} & +\frac{L_x}{2} \\ +c_\tau & -c_\tau & +c_\tau & -c_\tau \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}, \tag{2.11}$$

Equivalently,

$$\tau_x = \frac{L_y}{2}(-f_1 - f_2 + f_3 + f_4), \quad \tau_y = \frac{L_x}{2}(f_1 - f_2 - f_3 + f_4), \quad \tau_z = c_\tau(f_1 - f_2 + f_3 - f_4), \quad T = \sum_{i=1}^{4} f_i.$$

A common low-level model is $f_i = k_f\,\Omega_i^2$ where $\Omega_i$ is the motor speed. Eq. (2.11) then becomes a linear map

in $\Omega_i^2$

$$\begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = k_f \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\frac{L_y}{2} & -\frac{L_y}{2} & +\frac{L_y}{2} & +\frac{L_y}{2} \\ +\frac{L_x}{2} & -\frac{L_x}{2} & -\frac{L_x}{2} & +\frac{L_x}{2} \\ +c_\tau & -c_\tau & +c_\tau & -c_\tau \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}, \tag{2.12}$$

## 2.6 Hover Equilibrium

At a hovering steady state $\dot{\boldsymbol{x}} = \mathbf{0}$ with a fixed inertial position and constant yaw, we have

$$T_s = mg, \qquad \boldsymbol{\tau}_s = \mathbf{0}, \qquad f_{i,s} = \tfrac{mg}{4},\ i = 1, \ldots, 4. \tag{2.13}$$

The corresponding equilibrium state can be taken as

$$\boldsymbol{x}_s = \begin{bmatrix} \boldsymbol{p}_s \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \quad \psi_s \in \mathbb{R}, \tag{2.14}$$

which will be used for linearization and setpoint tracking.

## 2.7 Compact Nonlinear Model Summary

Collecting the above, the dynamics have the standard control–affine form

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}) = \begin{bmatrix} \dot{\boldsymbol{p}} \\ \dot{\boldsymbol{v}} \\ \dot{\boldsymbol{\eta}} \\ \dot{\boldsymbol{\omega}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{v} \\ g\,\boldsymbol{e}_3 - \dfrac{T}{m}\,\boldsymbol{b}_3(\phi, \theta, \psi) \\ W(\phi, \theta)\,\boldsymbol{\omega} \\ J^{-1}(\boldsymbol{\tau} - \boldsymbol{\omega} \times (J\,\boldsymbol{\omega})) \end{bmatrix}, \qquad \boldsymbol{u} = \begin{bmatrix} T \\ \boldsymbol{\tau} \end{bmatrix}, \ \ \boldsymbol{\tau} \text{ given by } \textit{Eq. (2.11)}. \tag{2.15}$$

Here $\boldsymbol{b}_3$ is the third column of $R(\phi, \theta, \psi) = R_z(\psi) R_y(\theta) R_x(\phi)$ and $W(\phi, \theta)$ is defined in Eq. (2.7). The model is well-defined for $\cos\theta \neq 0$.

## 2.8 Modeling Assumptions and Possible Extensions

This work neglects aerodynamic drag, ground effect, and propeller dynamics, and assumes $J$ diagonal and a rigid frame. These effects can be incorporated if needed via additive forces/torques or augmented states (e.g., first–order motor dynamics). For large attitude excursions or to avoid the Euler singularity at $\theta = \pm\frac{\pi}{2}$, a quaternion attitude representation is an alternative with identical force/torque equations.

# 3 Minimum Principle Formulation

This section derives the optimality system (state/costate ODEs and control law) using Pontryagin's Minimum Principle (PMP) for the quadrotor modeled in Sec. 2. From this point onward, we omit boldface for all system variables; only direction vectors (e.g., $\mathbf{e}_3, \mathbf{b}_3$) remain bold.

## 3.1 Dynamics and Cost

The standard quadrotor model can be written as

$$\dot{x} \;=\; f(x,u) \;=\; f_0(x) + G(x)\,u, \tag{3.1}$$

with

$$f_0(x) = \begin{bmatrix} v \\ g\,\boldsymbol{e}_3 \\ W(\eta)\,\omega \\ -\,J^{-1}(\,\omega \times (J\omega)\,) \end{bmatrix}, \qquad G(x) = \begin{bmatrix} \mathbf{0}_{3\times1} & \mathbf{0}_{3\times3} \\ -\frac{1}{m}\,\boldsymbol{b}_3(\eta) & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times1} & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times1} & J^{-1} \end{bmatrix}, \tag{3.2}$$

where $m > 0$ and $J \succ 0$ is the inertia matrix. The tracking cost is

$$J \;=\; \frac{1}{2}\,(x(t_f)-x_f)^\top Q_f\,(x(t_f)-x_f) + \int_0^{t_f} \left[\frac{1}{2}\,(x-x_r(t))^\top Q\,(x-x_r(t)) + \frac{1}{2}\,(u-u_r(t))^\top R\,(u-u_r(t))\right] dt, \tag{3.3}$$

with $Q \succeq 0$, $R \succ 0$, and arbitrary *time-varying* references $x_r(t)$ and $u_r(t)$.

## 3.2 Hamiltonian, Stationarity, and Boundary Conditions

Define the costate $\lambda(t) \in \mathbb{R}^{12}$ and the Hamiltonian

$$\mathcal{H}(x,u,\lambda,t) \;=\; \tfrac{1}{2}(x-x_r)^\top Q(x-x_r) + \tfrac{1}{2}(u-u_r)^\top R(u-u_r) + \lambda^\top\big(f_0(x) + G(x)u\big). \tag{3.4}$$

For fixed $t_f$, PMP gives the TPBVP:

$$\dot{x}^\star = \frac{\partial \mathcal{H}}{\partial \lambda}(x^\star,u^\star,\lambda^\star,t) \;=\; f_0(x^\star) + G(x^\star)u^\star, \tag{3.5}$$

$$\dot{\lambda}^\star = -\,\frac{\partial \mathcal{H}}{\partial x}(x^\star,u^\star,\lambda^\star,t), \tag{3.6}$$

$$u^\star(t) = \arg\min_{u\in\mathcal{U}}\; \mathcal{H}(x^\star,u,\lambda^\star,t), \tag{3.7}$$

$$x^\star(0) = x_0, \qquad \lambda^\star(t_f) = Q_f\big(x^\star(t_f) - x_f\big). \tag{3.8}$$

**Stationarity (unconstrained).** Since $\partial\mathcal{H}/\partial u = R(u-u_r) + G^\top\lambda$,

$$R\,(u^\star - u_r) + G(x^\star)^\top\lambda^\star = 0, \tag{3.9}$$

$$u^\star = u_r - R^{-1}G(x^\star)^\top\lambda^\star. \tag{3.10}$$

**With bounds.** For box constraints $\mathcal{U} = [u_{\min}, u_{\max}]$,

$$u^\star \;=\; \Pi_{\mathcal{U}}\Big(u_r - R^{-1}G(x^\star)^\top \lambda^\star\Big), \tag{3.11}$$

where $\Pi_{\mathcal{U}}$ is the elementwise projection (saturation).

## 3.3 Costate Equations

Partition the costate conformably with $x$:

$$\lambda \;=\; \begin{bmatrix}\lambda_p^\top & \lambda_v^\top & \lambda_\eta^\top & \lambda_\omega^\top\end{bmatrix}^\top. \tag{3.12}$$

Using Eq. (3.4) and $\dot{\lambda} = -\partial\mathcal{H}/\partial x$, one obtains

$$\dot{\lambda}_p = -Q_p\,(p - p_r), \tag{3.13}$$

$$\dot{\lambda}_v = -Q_v\,(v - v_r) \;-\; \lambda_p, \tag{3.14}$$

$$\dot{\lambda}_\eta = -Q_\eta\,(\eta - \eta_r) \;+\; \frac{T^\star}{m}\left(\frac{\partial \boldsymbol{b}_3(\eta)}{\partial \eta}\right)^\top \lambda_v \;-\; \left(\frac{\partial (W(\eta)\omega)}{\partial \eta}\right)^\top \lambda_\eta, \tag{3.15}$$

$$\dot{\lambda}_\omega = -Q_\omega\,(\omega - \omega_r) \;-\; W(\eta)^\top \lambda_\eta \;+\; \left(\frac{\partial}{\partial \omega}\big[\omega \times (J\omega)\big]\right)^\top J^{-\top}\lambda_\omega. \tag{3.16}$$

where for any constant $J$,

$$\frac{\partial}{\partial \omega}\big[\omega \times (J\omega)\big] \;=\; [J\omega]_\times - [\omega]_\times J, \quad [a]_\times := \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}. \tag{3.17}$$

and from Eq. (2.9), the columns of $\frac{\partial \boldsymbol{b}_3}{\partial \eta} \in \mathbb{R}^{3\times 3}$ are

$$\begin{aligned}
\frac{\partial \boldsymbol{b}_3}{\partial \phi} &= \begin{bmatrix} -\sin\phi\sin\theta\cos\psi + \sin\psi\cos\phi \\ -\sin\phi\sin\psi\sin\theta - \cos\phi\cos\psi \\ -\sin\phi\cos\theta \end{bmatrix}, \\
\frac{\partial \boldsymbol{b}_3}{\partial \theta} &= \begin{bmatrix} \cos\phi\cos\psi\cos\theta \\ \sin\psi\cos\phi\cos\theta \\ -\sin\theta\cos\phi \end{bmatrix}, \\
\frac{\partial \boldsymbol{b}_3}{\partial \psi} &= \begin{bmatrix} \sin\phi\cos\psi - \sin\psi\sin\theta\cos\phi \\ \sin\phi\sin\psi + \sin\theta\cos\phi\cos\psi \\ 0 \end{bmatrix}.
\end{aligned} \tag{3.18}$$

From Eq. (2.7), $W$ is independent of $\psi$. Hence

$$\frac{\partial (W(\eta)\omega)}{\partial \eta} \;=\; \left(\frac{\partial W}{\partial \phi}\right)\omega \;\Bigg|\; \left(\frac{\partial W}{\partial \theta}\right)\omega \;\Bigg|\; \mathbf{0}_{3\times 1}, \tag{3.19}$$

with

$$\frac{\partial W}{\partial \phi} = \begin{bmatrix} 0 & \cos\phi\,\tan\theta & -\sin\phi\,\tan\theta \\ 0 & -\sin\phi & -\cos\phi \\ 0 & \cos\phi/\cos\theta & -\sin\phi/\cos\theta \end{bmatrix},$$
$$\frac{\partial W}{\partial \theta} = \begin{bmatrix} 0 & \sin\phi\,\sec^2\theta & \cos\phi\,\sec^2\theta \\ 0 & 0 & 0 \\ 0 & \sin\phi\,\sec\theta\,\tan\theta & \cos\phi\,\sec\theta\,\tan\theta \end{bmatrix}. \tag{3.20}$$

### 3.4 Closed-Form Control Components

Let

$$R = \mathrm{diag}(r_T, R_\tau), \qquad R_\tau \in \mathbb{R}^{3\times 3},\ r_T > 0. \tag{3.21}$$

Using Eq. (3.10) and the structure of $G(x)$ in Eq. (3.2), the optimal inputs separate as

$$T^\star = T_r + \frac{1}{m\,r_T}\,\boldsymbol{b}_3(\eta)^\top \lambda_v, \tag{3.22}$$
$$\tau^\star = \tau_r - R_\tau^{-1} J^{-\top} \lambda_\omega, \tag{3.23}$$

followed by the projection Eq. (3.11) if bounds are present.

### 3.5 Boundary and Transversality Conditions

For fixed final time $t_f$ and quadratic terminal penalty Eq. (3.3),

$$\lambda(t_f) = Q_f\,(x(t_f) - x_f). \tag{3.24}$$

If some components of $x(t_f)$ are hard (fixed), the corresponding components of $\lambda(t_f)$ are free. If $t_f$ is free and the integrand is time-invariant, the transversality condition adds

$$\mathcal{H}\big(x^\star(t_f), u^\star(t_f), \lambda^\star(t_f)\big) = 0. \tag{3.25}$$

## 4 Linearization

This section linearizes the model about hover to obtain deviation dynamics $(A, B)$ and verifies controllability to justify subsequent linear designs.

### 4.1 Hovering Equilibrium

We linearize the nonlinear model Eq. (2.6)–Eq. (2.10) about the previously defined hover equilibrium $(x_s, u_s)$ (Section 2.6). Define deviations $\tilde{x} := x - x_s$, $\tilde{u} := u - u_s$, where $\delta\tilde{x} = \delta x$. The first-order Taylor expansion

gives

$$\dot{\tilde{x}} \;=\; A\,\tilde{x} + B\,\tilde{u}, \qquad A := \left.\frac{\partial f}{\partial x}\right|_{(x_s,u_s)}, \quad B := \left.\frac{\partial f}{\partial u}\right|_{(x_s,u_s)}. \tag{4.1}$$

and

$$\delta\dot{x} = \left.\frac{\partial f}{\partial x}\right|_{(x_s,u_s)} \delta x + \left.\frac{\partial f}{\partial u}\right|_{(x_s,u_s)} \delta u, \qquad \delta x = \begin{bmatrix} \delta p \\ \delta v \\ \delta \eta \\ \delta \omega \end{bmatrix}, \quad \delta u = \begin{bmatrix} \delta T \\ \delta \tau \end{bmatrix}. \tag{4.2}$$

$$\begin{aligned} &\delta\dot{p} = \delta v, \\ &\delta\dot{\eta} = W(\eta_s)\,\delta\omega \;+\; \delta W\,\omega_s \;=\; \delta\omega \quad \text{(at hover: } \eta_s = 0,\ \omega_s = 0). \end{aligned} \tag{4.3}$$

From Eq. (2.8) and Eq. (2.9), perturb around hover

$$\delta\dot{v} = -\frac{1}{m}\,\boldsymbol{b}_3(\eta_s)\,\delta T \;-\; \frac{T_s}{m}\left(\left.\frac{\partial \boldsymbol{b}_3}{\partial \eta}\right|_s\right)\delta\eta. \tag{4.4}$$

Using Eq. (2.9) and Eq. (3.18) at the hover attitude

$$\boldsymbol{b}_3(\eta_s) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad \left.\frac{\partial \boldsymbol{b}_3}{\partial \phi}\right|_s = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \qquad \left.\frac{\partial \boldsymbol{b}_3}{\partial \theta}\right|_s = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \left.\frac{\partial \boldsymbol{b}_3}{\partial \psi_s}\right|_s = 0, \tag{4.5}$$

and $T_s/m = g$, this reduces to

$$\delta\dot{v} = \begin{bmatrix} 0 & -g & 0 \\ g & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \delta\eta \;+\; \begin{bmatrix} 0 \\ 0 \\ -\frac{1}{m} \end{bmatrix} \delta T. \tag{4.6}$$

Linearizing Eq. (2.10) about $\omega_s = 0$

$$\begin{gathered} \delta\dot{\omega} = \delta\Big(-\,J^{-1}\big(\omega \times (J\omega)\big)\Big) + J^{-1}\delta\tau = -\,J^{-1}(\omega_s \times (J\,\delta\omega) + \delta\omega \times (J\,\omega_s)) + J^{-1}\delta\tau. \\ \delta\dot{\omega} = J^{-1}\delta\tau. \end{gathered} \tag{4.7}$$

and hence,

$$\begin{aligned} &\left.\frac{\partial \dot{p}}{\partial v}\right|_s = I_3, \\ &\left.\frac{\partial \dot{v}}{\partial \eta}\right|_s = \begin{bmatrix} 0 & -g & 0 \\ g & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \left.\frac{\partial \dot{v}}{\partial T}\right|_s = \begin{bmatrix} 0 \\ 0 \\ -\frac{1}{m} \end{bmatrix}, \\ &\left.\frac{\partial \dot{\eta}}{\partial \omega}\right|_s = I_3, \quad \left.\frac{\partial \dot{\omega}}{\partial \tau}\right|_s = J^{-1}. \end{aligned} \tag{4.8}$$

Substituting the Jacobians from Eq. (4.3)–Eq. (4.6), the linearized rotational block gives

$$A = \begin{bmatrix} 0_{3,3} & I_3 & 0_{3,3} & 0_{3,3} \\ 0_{3,3} & 0_{3,3} & \left.\frac{\partial \dot{v}}{\partial \eta}\right|_s & 0_{3,3} \\ 0_{3,3} & 0_{3,3} & 0_{3,3} & I_3 \\ 0_{3,3} & 0_{3,3} & 0_{3,3} & 0_{3,3} \end{bmatrix}, \qquad B = \begin{bmatrix} 0_{3,1} & 0_{3,3} \\ \left.\frac{\partial \dot{v}}{\partial T}\right|_s & 0_{3,3} \\ 0_{3,1} & 0_{3,3} \\ 0_{3,1} & J^{-1} \end{bmatrix}. \tag{4.9}$$

### 4.2 Controllability

Forming $C = [\, B \;\; AB \;\; A^2B \;\; \cdots \;\; A^{11}B \,]$ from Eq. (4.9) and finding $\operatorname{rank}(C)$ to be equal to 12 shows controllability.

## 5 Minimum Principle on the Linearized Model

We apply PMP to the hover–linearized model from Section 4:

$$\dot{\tilde{x}} = A\tilde{x} + B\tilde{u}, \qquad \tilde{x} := x - x_s, \;\; \tilde{u} := u - u_s, \tag{5.1}$$

where $A = \partial f/\partial x|_{(x_s,u_s)}$ and $B = \partial f/\partial u|_{(x_s,u_s)}$.

### 5.1 Finite-horizon quadratic problem

For fixed $t_f$, consider

$$J = \tfrac{1}{2}\,\tilde{x}(t_f)^\top Q_f\,\tilde{x}(t_f) + \tfrac{1}{2}\int_0^{t_f} \left(\tilde{x}^\top Q\,\tilde{x} + \tilde{u}^\top R\,\tilde{u}\right) dt, \qquad Q \succeq 0,\; R \succ 0. \tag{5.2}$$

Define the Hamiltonian

$$H(\tilde{x}, \tilde{u}, \lambda) = \tfrac{1}{2}\tilde{x}^\top Q\tilde{x} + \tfrac{1}{2}\tilde{u}^\top R\tilde{u} + \lambda^\top(A\tilde{x} + B\tilde{u}). \tag{5.3}$$

PMP gives the forward–backward system

$$\dot{\tilde{x}}^\star = \frac{\partial H}{\partial \lambda} = A\tilde{x}^\star + B\tilde{u}^\star, \qquad \tilde{x}^\star(0) = \tilde{x}_0, \tag{5.4}$$

$$\dot{\lambda}^\star = -\frac{\partial H}{\partial \tilde{x}} = -Q\tilde{x}^\star - A^\top\lambda^\star, \qquad \text{(terminal BCs listed below)} \tag{5.5}$$

$$0 = \frac{\partial H}{\partial \tilde{u}} = R\tilde{u}^\star + B^\top\lambda^\star \qquad \Longrightarrow \qquad \tilde{u}^\star = -R^{-1}B^\top\lambda^\star. \tag{5.6}$$

**Terminal/transversality conditions.** Let $\Phi(\tilde{x}, t) = \frac{1}{2}\,\tilde{x}^\top Q_f\tilde{x}$ be the terminal cost and let $\Psi(\tilde{x}, t) = 0$ denote terminal equality constraints (possibly none).

- **Fixed terminal time $t_f$.**

- *Free terminal state (no constraints).*

$$\lambda^\star(t_f) = Q_f\,\tilde{x}^\star(t_f).$$

- *Terminal state fully specified* $\big(\tilde{x}^\star(t_f) = \tilde{x}_f\big)$.

$$\lambda^\star(t_f) \text{ is free} \quad (\text{equivalently, } \lambda^\star(t_f) = Q_f\tilde{x}^\star(t_f) + \nu \text{ with unrestricted } \nu).$$

- *Terminal manifold constraint* $\big(\Psi(\tilde{x}^\star(t_f)) = 0\big)$.

$$\lambda^\star(t_f) = Q_f\,\tilde{x}^\star(t_f) + \Psi_{\tilde{x}}^\top \nu,$$

  with multipliers $\nu$ chosen so that the constraint is satisfied.

- **Free terminal time $t_f$.**
  - *No terminal constraints.*

$$\lambda^\star(t_f) = Q_f\,\tilde{x}^\star(t_f), \qquad H^\star(t_f) = 0 \;\; (\partial\Phi/\partial t \equiv 0).$$

  - *With terminal constraints* $\big(\Psi(\tilde{x}^\star(t_f), t_f) = 0\big)$.

$$\lambda^\star(t_f) = Q_f\,\tilde{x}^\star(t_f) + \Psi_{\tilde{x}}^\top \nu, \qquad H^\star(t_f) = -\,\nu^\top \Psi_t.$$

where $\Psi_{\tilde{x}}$ is the Jacobian of $\Psi$ w.r.t. $\tilde{x}$, and $\Psi_t$ is the partial derivative w.r.t. $t$, both evaluated at $(\tilde{x}^\star(t_f), t_f)$.

**Numerical solution.** Equations Eq. (5.4)–Eq. (5.6) with the appropriate terminal conditions above define a TPBVP on $[0, t_f]$. In practice one uses a shooting or forward–backward sweep: guess the unknown multipliers (e.g., $\lambda^\star(t_f)$ and any $\nu$), integrate Eq. (5.5) backward and Eq. (5.4) forward with Eq. (5.6), and iterate until the boundary conditions are met.

## 5.2 Actuator limits

Assume elementwise bounds on the *physical* input $u$:

$$u_{\min} \le u(t) \le u_{\max} \quad \Longleftrightarrow \quad \tilde{u}_{\min} := u_{\min} - u_s \;\le\; \tilde{u}(t) \;\le\; \tilde{u}_{\max} := u_{\max} - u_s. \tag{5.7}$$

At each $t$, minimizing $H$ over this hyper-rectangle decouples by component and yields the closed-form projection:

$$\tilde{u}^\star(t) = \Pi_{[\tilde{u}_{\min},\,\tilde{u}_{\max}]}\Big(-R^{-1}B^\top\lambda^\star(t)\Big), \tag{5.8}$$

$$\tilde{u}_i^\star(t) = \min\Big\{\max\big\{-[R^{-1}B^\top\lambda^\star]_i,\; \tilde{u}_{\min,i}\big\},\; \tilde{u}_{\max,i}\Big\}. \tag{5.9}$$

In physical inputs:

$$u_i^\star(t) = \min\Big\{\max\big\{u_{s,i} - [R^{-1}B^\top\lambda^\star]_i,\; u_{\min,i}\big\},\; u_{\max,i}\Big\}. \tag{5.10}$$

*KKT check.* The clipping law Eq. (5.8) is equivalent to the KKT system $R\tilde{u}^\star + B^\top\lambda^\star + \mu^+ - \mu^- = 0$ with multipliers $\mu_i^\pm \geq 0$ and complementary slackness $\mu_i^+(\tilde{u}_i^\star - \tilde{u}_{\max,i}) = 0,\ \mu_i^-(\tilde{u}_{\min,i} - \tilde{u}_i^\star) = 0$.

### 5.3 Minimum-time variant (bang–bang)

For time-optimal regulation to a target (e.g., $\tilde{x}^\star(t_f) = 0$) with free $t_f$ and bounds $\tilde{u}_{\min} \leq \tilde{u} \leq \tilde{u}_{\max}$, the cost is $J = t_f = \int_0^{t_f} 1\,dt$ and

$$H(\tilde{x}, \tilde{u}, \lambda) = 1 + \lambda^\top(A\tilde{x} + B\tilde{u}), \qquad \dot{\lambda} = -A^\top\lambda. \tag{5.11}$$

With free terminal time, $H^\star(t_f) = 0$ (normal extremals). Minimizing $H$ over the box reduces to a linear program in $\tilde{u}$ with switching function $\sigma(t) := B^\top\lambda(t)$, giving componentwise bang–bang control:

$$\tilde{u}_i^\star(t) = \begin{cases} \tilde{u}_{\min,i}, & \sigma_i(t) > 0, \\ \tilde{u}_{\max,i}, & \sigma_i(t) < 0, \\ \text{any in } [\tilde{u}_{\min,i}, \tilde{u}_{\max,i}], & \sigma_i(t) = 0 \text{ (singular)}. \end{cases} \tag{5.12}$$

A compact equivalent is

$$\tilde{u}^\star(t) = \tfrac{1}{2}(\tilde{u}_{\max} + \tilde{u}_{\min}) - \tfrac{1}{2}\,\mathrm{Diag}(\tilde{u}_{\max} - \tilde{u}_{\min})\,\mathrm{sign}\big(\sigma(t)\big), \tag{5.13}$$

with any selection $\mathrm{sign}(0) \in [-1, 1]$. In physical coordinates,

$$u^\star(t) = \tfrac{1}{2}(u_{\max} + u_{\min}) - \tfrac{1}{2}\,\mathrm{Diag}(u_{\max} - u_{\min})\,\mathrm{sign}\big(B^\top\lambda(t)\big). \tag{5.14}$$

*Notes.* (i) Switching times are zeros of $\sigma_i(t)$ and are solved as part of the TPBVP; (ii) true singular arcs ($\sigma_i \equiv 0$ on an interval) require higher-order conditions and are nongeneric for controllable LTI systems.

### 5.4 Infinite-horizon case: boundary condition and adjusted TPBVP

On $[0, \infty)$ with the quadratic running cost in Eq. (5.2) (no terminal penalty), PMP keeps the same state/costate ODEs and stationarity law, but the boundary at infinity replaces terminal conditions:

$$\dot{\tilde{x}}^\star = A\tilde{x}^\star + B\tilde{u}^\star, \qquad \dot{\lambda}^\star = -Q\tilde{x}^\star - A^\top\lambda^\star, \qquad R\tilde{u}^\star + B^\top\lambda^\star = 0, \tag{5.15}$$

$$\tilde{x}^\star(0) = \tilde{x}_0, \qquad \lim_{t\to\infty}\lambda^\star(t) = \mathbf{0}, \qquad \text{and} \quad \lim_{t\to\infty} H^\star(t) = 0 \ \text{ if terminal time is free.} \tag{5.16}$$

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## 6 Linear–Quadratic Regulator (LQR)

In this section we design a state-feedback regulator for the hover–linearized deviation model

$$\dot{\tilde{x}} = A\tilde{x} + B\tilde{u}, \qquad \tilde{x} := x - x_s, \quad \tilde{u} := u - u_s, \tag{6.1}$$

with symmetric weights $Q \succeq 0$ and $R \succ 0$.

### 6.1 Finite-horizon LQR

For a fixed horizon $[0, t_f]$, consider

$$J = \tfrac{1}{2}\,\tilde{x}(t_f)^\top Q_f \tilde{x}(t_f) + \tfrac{1}{2}\int_0^{t_f} \left(\tilde{x}^\top Q\,\tilde{x} + \tilde{u}^\top R\,\tilde{u}\right) dt. \tag{6.2}$$

The optimal value has quadratic form $V(t, \tilde{x}) = \frac{1}{2}\,\tilde{x}^\top P(t)\tilde{x}$ with

$$-\dot{P}(t) = A^\top P + PA - PBR^{-1}B^\top P + Q, \qquad P(t_f) = Q_f. \tag{6.3}$$

In practice, $P(t)$ is obtained by integrating Eq. (6.3) backward from $t_f$ to 0, or equivalently by solving the coupled state–costate TPBVP. The optimal control and closed-loop dynamics are

$$\tilde{u}^\star(t) = -K(t)\tilde{x}(t), \qquad K(t) := R^{-1}B^\top P(t), \qquad \dot{\tilde{x}} = (A - BK(t))\tilde{x}. \tag{6.4}$$

In original variables,

$$u^\star(t) = u_s - K(t)\big(x(t) - x_s\big). \tag{6.5}$$

The boundary-value problem Eq. (6.3) with terminal condition $Q_f \succeq 0$ has a symmetric solution $P(t)$ on $[0, t_f]$. The optimal cost is $J^\star = \frac{1}{2}\,\tilde{x}(0)^\top P(0)\tilde{x}(0)$.

### 6.2 Infinite-horizon LQR

On $[0, \infty)$ with running cost $\int_0^\infty (\tilde{x}^\top Q\tilde{x} + \tilde{u}^\top R\tilde{u})\,dt$, the optimal feedback is constant. There exists a unique symmetric, positive semidefinite *stabilizing* solution $P$ of the algebraic Riccati equation

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0. \tag{6.6}$$

The optimal controller, closed loop, and value function are

$$\tilde{u}^\star = -K\tilde{x}, \qquad K := R^{-1}B^\top P, \qquad \dot{\tilde{x}} = (A - BK)\tilde{x}, \qquad J^\star = \tfrac{1}{2}\,\tilde{x}(0)^\top P\,\tilde{x}(0). \tag{6.7}$$

Implementation in original variables:

$$u^\star(t) = u_s - K\big(x(t) - x_s\big). \tag{6.8}$$

In general, with actuator limits, the implemented law is the saturated feedback $\tilde{u} = \Pi_{[\tilde{u}_{\min}, \tilde{u}_{\max}]}(-K\tilde{x})$; stability then holds in a region around the origin determined by the bounds. In other words, LQR is solved unconstrained, yet actuator limits are enforced after the control law as an input saturation.

## 7 Model Predictive Control

We formulate a finite-horizon optimal control problem on a discrete grid $t_k = k\,\Delta t$, $k = 0, 1, \ldots$, with sampling time $\Delta t > 0$. At each $t_0$, the problem is solved, the first input is applied, and the horizon is shifted forward (receding horizon).

## 7.1 Linear MPC

From Section 4, about $(x_s, u_s)$ we have

$$\dot{\tilde{x}} \;=\; A\,\tilde{x} + B\,\tilde{u}, \qquad \tilde{x} = \begin{bmatrix} \tilde{p} \\ \tilde{v} \\ \tilde{\eta} \\ \tilde{\omega} \end{bmatrix}, \;\; \tilde{u} = \begin{bmatrix} \tilde{T} \\ \tilde{\tau} \end{bmatrix}, \tag{7.1}$$

with the block structure derived in Eq. (4.9). In particular, at hover

$$\left.\frac{\partial \dot{v}}{\partial \eta}\right|_s = \underbrace{\begin{bmatrix} 0 & -g & 0 \\ g & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{A_{v\eta}}, \qquad \left.\frac{\partial \dot{v}}{\partial T}\right|_s = \underbrace{\begin{bmatrix} 0 \\ 0 \\ -\frac{1}{m} \end{bmatrix}}_{b_T}, \qquad \left.\frac{\partial \dot{\eta}}{\partial \omega}\right|_s = I_3, \qquad \left.\frac{\partial \dot{\omega}}{\partial \tau}\right|_s = J^{-1}. \tag{7.2}$$

**ZOH discretization.** Holding $\tilde{u}$ constant on $[t_k, t_{k+1})$,

$$\tilde{x}_{k+1} = A_d\,\tilde{x}_k + B_d\,\tilde{u}_k, \tag{7.3a}$$

$$A_d = \mathrm{e}^{A\Delta t}, \tag{7.3b}$$

$$B_d = \int_0^{\Delta t} \mathrm{e}^{A\tau}\,B\,d\tau. \tag{7.3c}$$

*Convenient constructions for $B_d$:*

$$B_d = A^{-1}(A_d - I)\,B, \qquad \text{if } A \text{ is nonsingular}, \tag{7.4}$$

$$\mathrm{e}^{\begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}\Delta t} = \begin{bmatrix} A_d & B_d \\ 0 & I \end{bmatrix}, \qquad \text{always}. \tag{7.5}$$

**Discrete updates.** From the series form $A_d = I + \Delta t A + \frac{\Delta t^2}{2}A^2 + \cdots$ and $B_d = \Delta t\,B + \frac{\Delta t^2}{2}AB + \cdots$, and the blocks in Eqs. (4.6) and (4.9), the first nontrivial discrete terms are

$$\tilde{p}_{k+1} = \tilde{p}_k + \Delta t\,\tilde{v}_k + \frac{\Delta t^2}{2}\Big(A_{v\eta}\,\tilde{\eta}_k \;+\; b_T\,\tilde{T}_k\Big) \;+\; O(\Delta t^3), \tag{7.6}$$

$$\tilde{v}_{k+1} = \tilde{v}_k + \Delta t\Big(A_{v\eta}\,\tilde{\eta}_k \;+\; b_T\,\tilde{T}_k\Big) + \frac{\Delta t^2}{2}\,A_{v\eta}\,\tilde{\omega}_k \;+\; O(\Delta t^2), \tag{7.7}$$

$$\tilde{\eta}_{k+1} = \tilde{\eta}_k + \Delta t\,\tilde{\omega}_k \;+\; O(\Delta t^2), \tag{7.8}$$

$$\tilde{\omega}_{k+1} = \tilde{\omega}_k + \Delta t\,J^{-1}\tilde{\tau}_k \;+\; O(\Delta t^2). \tag{7.9}$$

Equations Eq. (7.6)–Eq. (7.9) make the discrete couplings explicit in the same detailed variables as Section 4. In implementation, the exact $A_d, B_d$ from Eq. (7.3) are used.

**Cost, input limits, and problem statement.** Discretizing the continuous cost introduced in Section 3,

$$J_N \;=\; \tfrac{1}{2}\,(x_N - x_r(t_N))^\top Q_f\,(x_N - x_r(t_N)) \;+\; \sum_{k=0}^{N-1}\Big[\tfrac{1}{2}\,(x_k - x_r(t_k))^\top Q\,(x_k - x_r(t_k)) + \tfrac{1}{2}\,(u_k - u_r(t_k))^\top R\,(u_k - u_r(t_k))\Big]\,\Delta t_{\text{ctrl}}. \tag{7.10}$$

with *input limits*:

$$u_{\min} \;\le\; u_k \;\le\; u_{\max}, \qquad |u_k - u_{k-1}| \le \Delta u_{\max}. \tag{7.11}$$

Set $Q_f = P$, where $P$ solves the discrete-time DARE for $(A_d, B_d, Q, R)$ (LQR terminal weight). The resulting finite-horizon problem (quadratic cost + linear dynamics/limits) is a convex quadratic program (QP). At each $t_0$ the problem is solved, $u_0^\star$ is applied, the horizon is shifted, and the procedure repeats.

## 7.2 Nonlinear MPC

The continuous dynamics of Section 2 are discretized with the classical fourth–order Runge–Kutta (RK4) method. Over each sampling interval $[t_k, t_{k+1})$ the input is held constant, $u(t) \equiv u_k$, and the vector field $f(x,u) = f_0(x) + G(x)u$ is evaluated at four stages: $k_1 = f(x_k, u_k)$, $k_2 = f\big(x_k + \frac{\Delta t}{2} k_1,\, u_k\big)$, $k_3 = f\big(x_k + \frac{\Delta t}{2} k_2,\, u_k\big)$, $k_4 = f\big(x_k + \Delta t\, k_3,\, u_k\big)$. The state is advanced by $x_{k+1} = x_k + \frac{\Delta t}{6}\,(k_1 + 2k_2 + 2k_3 + k_4)$, which defines the one–step map $x_{k+1} = f_d(x_k, u_k)$. This explicit integrator offers high accuracy for smooth rigid–body dynamics (local truncation error $O(\Delta t^5)$, global error $O(\Delta t^4)$) while remaining simple to implement. Each stage evaluation uses the expressions in Eqs. (2.7), (2.9) and (2.10).

$$k_1 = \begin{bmatrix} v_k \\ g\,\boldsymbol{e}_3 - \dfrac{T_k}{m}\,\boldsymbol{b}_3(\eta_k) \\ W(\eta_k)\,\omega_k \\ -\,J^{-1}\big(\omega_k \times (J\omega_k)\big) \;+\; J^{-1}\tau_k \end{bmatrix}, \tag{7.12a}$$

$$k_2 = f\Big(x_k + \tfrac{\Delta t}{2}\,k_1,\, u_k\Big), \tag{7.12b}$$

$$k_3 = f\Big(x_k + \tfrac{\Delta t}{2}\,k_2,\, u_k\Big), \tag{7.12c}$$

$$k_4 = f\Big(x_k + \Delta t\,k_3,\, u_k\Big), \tag{7.12d}$$

$$x_{k+1} = x_k \;+\; \tfrac{\Delta t}{6}\Big(k_1 + 2k_2 + 2k_3 + k_4\Big). \tag{7.12e}$$

Each $k_i \in \mathbb{R}^{12}$ is partitioned as $k_i = \begin{bmatrix} k_{i,p}^\top & k_{i,v}^\top & k_{i,\eta}^\top & k_{i,\omega}^\top \end{bmatrix}^\top$. Written componentwise,

$$p_{k+1} = p_k + \tfrac{\Delta t}{6}\,(k_{1,p} + 2k_{2,p} + 2k_{3,p} + k_{4,p}), \tag{7.13}$$

$$v_{k+1} = v_k + \tfrac{\Delta t}{6}\,(k_{1,v} + 2k_{2,v} + 2k_{3,v} + k_{4,v}), \tag{7.14}$$

$$\eta_{k+1} = \eta_k + \tfrac{\Delta t}{6}\,(k_{1,\eta} + 2k_{2,\eta} + 2k_{3,\eta} + k_{4,\eta}), \tag{7.15}$$

$$\omega_{k+1} = \omega_k + \tfrac{\Delta t}{6}\,(k_{1,\omega} + 2k_{2,\omega} + 2k_{3,\omega} + k_{4,\omega}). \tag{7.16}$$

**Cost, input limits, and problem statement.** Use the same quadratic tracking cost Eq. (7.10) and the same

input limits Eq. (7.11). The finite-horizon problem is

$$\min_{\{x_k,u_k\}_{k=0}^{N-1},\, x_N} \quad J_N$$

$$\text{s.t.} \quad x_{k+1} - f_d(x_k, u_k) = 0 \;\text{ with } f_d \text{ given by Eq. (7.13)–Eq. (7.16)},\; k = 0, \ldots, N-1, \tag{7.17}$$

$$u_{\min} \le u_k \le u_{\max}, \quad |u_k - u_{k-1}| \le \Delta u_{\max} \text{ (optional)}, \tag{7.18}$$

$$x_0 \text{ given.} \tag{7.19}$$

This yields a sparse nonlinear program (NLP) (multiple shooting with the explicit RK4 equalities). At each $t_0$ the problem is solved, $u_0^\star$ is applied, the horizon is shifted, and the procedure repeats.

# 8 Simulations and Results

This section compares LQR (on linearized and nonlinear plants), linear and nonlinear MPC, and the PMP/Ipopt controller on the nonlinear model, reporting total costs and 2%-band settling, overshoot and peak values for the zero-referenced states.

## 8.1 System Parameters

Table 1 lists the geometry and mass properties used in all simulations, along with the derived hover thrust numbers used to define the operating point. These are the parameters of the actual Quanser QDrone2. All simulations were run for 8 seconds, which was enough for reaching the full steady state.

**Table 1:** QDrone2 parameters and simulation settings.

| Quantity | Symbol | Value |
|---|---|---|
| Pitch arm length | $L_x$ | 0.2032 m |
| Roll arm length | $L_y$ | 0.254 m |
| Vehicle mass (takeoff) | $m$ | 1.504 kg |
| Roll Moment of Inertia | $J_x$ | 0.0220 kg m$^2$ |
| Pitch Moment of Inertia | $J_y$ | 0.0324 kg m$^2$ |
| Yaw Moment of Inertia | $J_z$ | 0.0544 kg m$^2$ |
| Gravity | $g$ | 9.81 m/s$^2$ |
| Total hover thrust | $T_s = mg$ | 14.754 N |
| Per-motor hover thrust | $f_i = mg/4$ | 3.689 N |
| Thrust coefficient | $k_f$ | $2.142\times10^{-8}$ N/RPM$^2$ |
| Motor torque constant | $c_\tau$ | 0.01451 N m/N |
| Maximum motor speed | $\Omega_{\max}$ | $2\times10^4$ RPM |
| Minimum motor speed | $\Omega_{\min}$ | 0 RPM |
| Initial states | $x_0$ | $\mathbf{0}$ |
| Reference position | $p_r$ | $\left[\,1,\, -1,\, -3\,\right]^\top$ m |
| Reference yaw | $\psi_r$ | 0.05 rad |
| Reference state | $x_r$ | $\left[\,p_r^\top,\, \mathbf{0}_3^\top,\, (0,0,\psi_r),\, \mathbf{0}_3^\top\,\right]^\top$ |

## 8.2 The dynamic model

By substituting in Eq. (2.15), the dynamic model is

$$\dot{p} = v, \tag{8.1a}$$

$$\dot{v} = \begin{bmatrix} -0.6649\,T\left(\sin\phi\,\sin\psi + \sin\theta\,\cos\phi\,\cos\psi\right) \\ -0.6649\,T\left(-\sin\phi\,\cos\psi + \sin\psi\,\sin\theta\,\cos\phi\right) \\ -0.6649\,T\left(\cos\phi\,\cos\theta\right) + 9.81 \end{bmatrix}, \tag{8.1b}$$

$$\dot{\eta} = \omega, \tag{8.1c}$$

$$\dot{\omega} = \begin{bmatrix} 45.4545\,\tau_x - \omega_y\omega_z \\ 30.8642\,\tau_y + \omega_x\omega_z \\ 18.3824\,\tau_z - 0.1912\,\omega_x\omega_y \end{bmatrix}. \tag{8.1d}$$

By substituting in (2.11), the motor mapping matrix is

$$\begin{aligned} \begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} &= \begin{bmatrix} 2.142\times10^{-8} & 2.142\times10^{-8} & 2.142\times10^{-8} & 2.142\times10^{-8} \\ -2.720\times10^{-9} & -2.720\times10^{-9} & 2.720\times10^{-9} & 2.720\times10^{-9} \\ 2.176\times10^{-9} & -2.176\times10^{-9} & -2.176\times10^{-9} & 2.176\times10^{-9} \\ 3.108\times10^{-10} & -3.108\times10^{-10} & 3.108\times10^{-10} & -3.108\times10^{-10} \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}, \\ \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix} &= \begin{bmatrix} 1.167\times10^{7} & -9.190\times10^{7} & 1.149\times10^{8} & 8.044\times10^{8} \\ 1.167\times10^{7} & -9.190\times10^{7} & -1.149\times10^{8} & -8.044\times10^{8} \\ 1.167\times10^{7} & 9.190\times10^{7} & -1.149\times10^{8} & 8.044\times10^{8} \\ 1.167\times10^{7} & 9.190\times10^{7} & 1.149\times10^{8} & -8.044\times10^{8} \end{bmatrix} \begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix}. \end{aligned} \tag{8.2}$$

and the absolute input bounds $u \in [u_{\min}, u_{\max}]$ come from the RPM box $\Omega_i \in [0, 2\times10^4]$ RPM via the map given in Eq. (8.2), yielding

$$u_{\min} = \begin{bmatrix} 0 \\ -2.176 \\ -1.7408 \\ -0.24864 \end{bmatrix}, \qquad u_{\max} = \begin{bmatrix} 34.272 \\ 2.176 \\ 1.7408 \\ 0.24864 \end{bmatrix} \quad (\mathrm{N}, \mathrm{N\,m}). \tag{8.3}$$

## 8.3 Linearized Model at Hover

We linearize about

$$x_s = \mathbf{0}, \qquad u_s = \begin{bmatrix} mg & 0 & 0 & 0 \end{bmatrix}^\top = \begin{bmatrix} 14.754 & 0 & 0 & 0 \end{bmatrix}^\top. \tag{8.4}$$

The corresponding motor thrusts at hover are $f_i = 3.689$ N, $i = 1..4$.

The linearized model is as follows:

$$A = \begin{bmatrix} \mathbf{0}_{3\times3} & I_3 & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times3} \\ & & 0 & -9.81 & 0 & \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & 9.81 & 0 & 0 & \mathbf{0}_{3\times3} \\ & & 0 & 0 & 0 & \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times3} & & I_3 \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times3} \end{bmatrix} \quad B = \begin{bmatrix} \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & \\ 0 & & & \\ 0 & & \mathbf{0}_{3\times3} & \\ -0.665 & & & \\ \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & \\ & 45.454 & 0 & 0 \\ \mathbf{0}_{3\times1} & 0 & 30.864 & 0 \\ & 0 & 0 & 18.382 \end{bmatrix} \tag{8.5}$$

.

The controllability matrix can be calculated as: formally, $C = [\, B \;\; AB \;\; A^2B \;\; \ldots \;\; A^{11}B \,]$ for $n = 12$. However, $A$ is strictly upper block–triangular (nilpotent) with $A^4 = \mathbf{0}$. Hence $A^kB = \mathbf{0}$ for all $k \geq 4$, and

$$C = \begin{bmatrix} B & AB & A^2B & A^3B \end{bmatrix}. \tag{8.6}$$

and,

$$C = \begin{bmatrix} \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & & \begin{matrix} 0 \\ 0 \\ -0.66489 \end{matrix} & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times1} & \begin{matrix} 0 \\ -445.909 \\ 0 \end{matrix} & \begin{matrix} 302.778 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 0 \end{matrix} \\ \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times1} & \begin{matrix} 0 \\ -445.909 \\ 0 \end{matrix} & \begin{matrix} 302.778 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 0 \end{matrix} & \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & \\ \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times1} & \begin{matrix} 45.455 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 30.864 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 18.382 \end{matrix} & \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & \\ \mathbf{0}_{3\times1} & \begin{matrix} 45.455 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 30.864 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 18.382 \end{matrix} & \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times1} & & \mathbf{0}_{3\times3} & \end{bmatrix}$$

The controllability matrix has full $\operatorname{rank}(C) = 12$.

## 8.4 LQR

We solve the continuous–time algebraic Riccati equation (CARE) for the hover linearization. The diagonal weights used are

$$\begin{aligned} \operatorname{diag}(Q) &= \left[\, 10.0,\; 10.0,\; 12.0,\; 2.0,\; 2.0,\; 2.0,\; 8.0,\; 8.0,\; 2.0,\; 0.5,\; 0.5,\; 0.5 \,\right], \\ \operatorname{diag}(R) &= \left[\, 1.0,\; 0.05,\; 0.05,\; 0.02 \,\right]. \end{aligned} \tag{8.7}$$

The open–loop eigenvalues of $A$ are

$$\lambda(A) = \underbrace{0, \ldots, 0}_{12}$$

and the closed–loop eigenvalues with $K_\infty$ are

$$\lambda(A-BK_\infty) = \begin{bmatrix} -105.3334 \\ -105.3334 \\ -83.3093 \\ -3.3967 + 2.5501\mathrm{j} \\ -3.3967 - 2.5501\mathrm{j} \\ -3.3967 + 2.5501\mathrm{j} \\ -3.3967 - 2.5501\mathrm{j} \\ -2.4327 \\ -2.4327 \\ -2.0006 \\ -1.1712 + 0.9644\mathrm{j} \\ -1.1712 - 0.9644\mathrm{j} \end{bmatrix}.$$

$$K_\infty = \begin{bmatrix} 0.000 & 0.000 & -3.464 & 0.000 & 0.000 & -3.525 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 14.142 & 0.000 & 0.000 & 11.273 & 0.000 & 30.192 & 0.000 & 0.000 & 3.437 & 0.000 & 0.000 \\ -14.142 & 0.000 & 0.000 & -11.273 & 0.000 & 0.000 & 0.000 & 30.192 & 0.000 & 0.000 & 3.437 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 10.000 & 0.000 & 0.000 & 5.119 \end{bmatrix}, \tag{8.8}$$

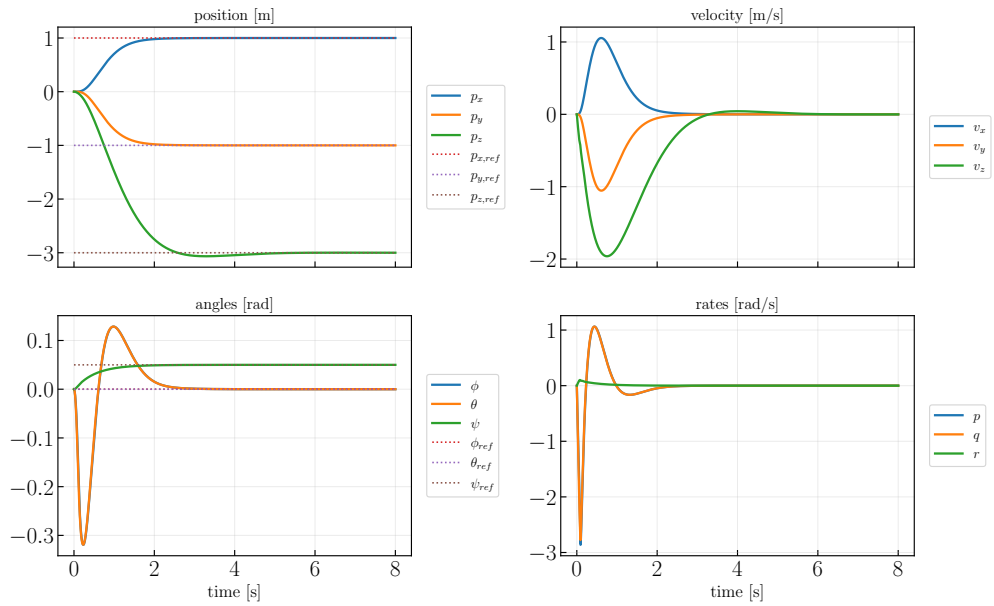
$$P_\infty = \begin{bmatrix} 7.971 & 0.000 & 0.000 & 2.177 & 0.000 & 0.000 & 0.000 & -2.430 & 0.000 & 0.000 & -0.021 & 0.000 \\ 0.000 & 7.971 & 0.000 & 0.000 & 2.177 & 0.000 & 2.430 & 0.000 & 0.000 & 0.021 & 0.000 & 0.000 \\ 0.000 & 0.000 & 12.212 & 0.000 & 0.000 & 5.213 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 2.177 & 0.000 & 0.000 & 1.488 & 0.000 & 0.000 & 0.000 & -1.916 & 0.000 & 0.000 & -0.017 & 0.000 \\ 0.000 & 2.177 & 0.000 & 0.000 & 1.488 & 0.000 & 1.916 & 0.000 & 0.000 & 0.017 & 0.000 & 0.000 \\ 0.000 & 0.000 & 5.213 & 0.000 & 0.000 & 5.305 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 2.430 & 0.000 & 0.000 & 1.916 & 0.000 & 5.022 & 0.000 & 0.000 & 0.045 & 0.000 & 0.000 \\ -2.430 & 0.000 & 0.000 & -1.916 & 0.000 & 0.000 & 0.000 & 5.022 & 0.000 & 0.000 & 0.045 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 1.024 & 0.000 & 0.000 & 0.012 \\ 0.000 & 0.021 & 0.000 & 0.000 & 0.017 & 0.000 & 0.045 & 0.000 & 0.000 & 0.005 & 0.000 & 0.000 \\ -0.021 & 0.000 & 0.000 & -0.017 & 0.000 & 0.000 & 0.000 & 0.045 & 0.000 & 0.000 & 0.005 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.012 & 0.000 & 0.000 & 0.006 \end{bmatrix} \tag{8.9}$$

and, $\lambda_{\min}(P_\infty) = 4.746\times10^{-3} > 0$, confirming positive definiteness.

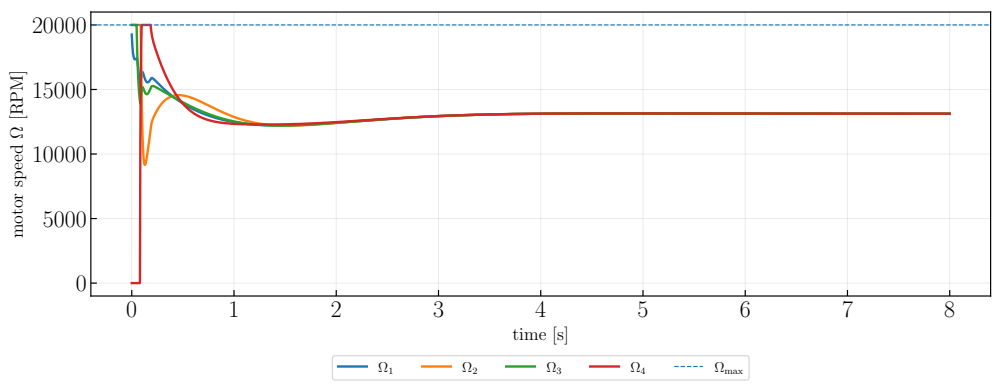
The optimal infinite–horizon gain $K_\infty$ was applied to both the linearized plant and the full nonlinear model. In both cases the closed loop is stable and the states converge to the commanded step in $[p_x, p_y, p_z, \psi]$ with essentially zero steady–state error (Figs. 3 and 4).

On the linearized plant, the quadratic objective is $J = 63.62$ with input saturation on 89 of 4001 steps. Transient settling time and overshoot (2% band) are: $p_x$ has $T_s = 2.004$ s and OS = 0.00%, $p_y$ $T_s = 2.008$ s and OS = 0.00%, $p_z$ $T_s = 3.600$ s and OS = 2.19%, $\psi$ is fast and non-oscillatory ($T_s = 1.976$ s, OS = 0.00%). Because $\phi$ and $\theta$ start and end at zero, overshoot is not informative; we instead report peak magnitudes: $|\phi|_{\mathrm{pk}} \approx 0.319$ rad, $|\theta|_{\mathrm{pk}} \approx 0.319$ rad.

Applying the same $K_\infty$ to the nonlinear plant yields $J = 67.15$ with input saturation on 109 of 4001 steps. Translations settle faster in $p_x$ and $p_y$ ($T_s = 1.826, 1.696$ s) with negligible overshoot (only $p_y$ shows 0.56%), the vertical channel is quicker yet still modestly underdamped ($p_z$ $T_s = 2.528$ s, OS = 1.95%), and yaw becomes slower and slightly underdamped ($\psi$ $T_s = 2.706$ s, OS = 16.40%). For attitudes we again report peaks: $|\phi|_{\mathrm{pk}} \approx 0.319$ rad, $|\theta|_{\mathrm{pk}} \approx 0.307$ rad.




**(a)** States.



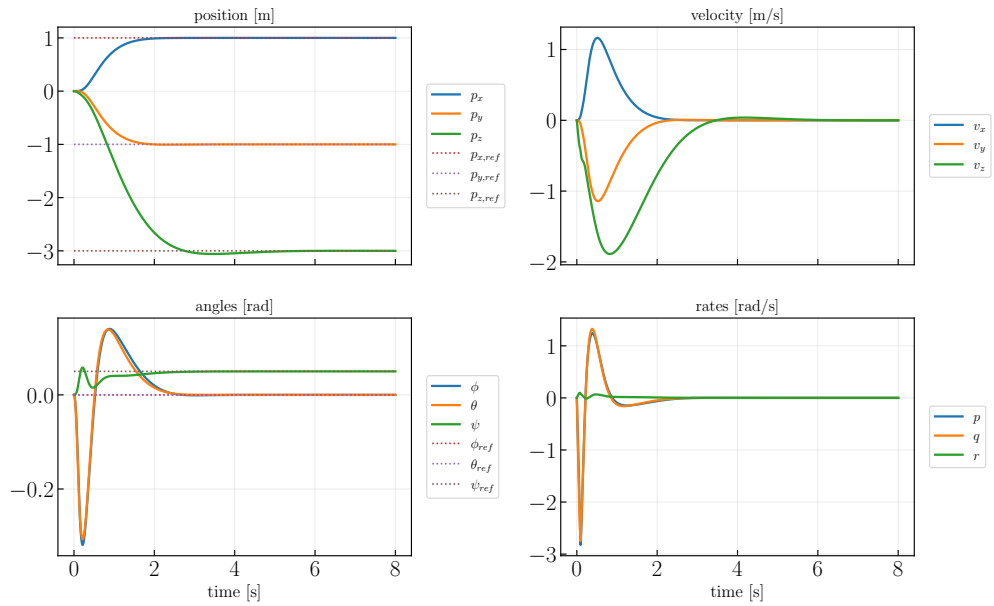

**(b)** Motor speeds $\Omega_i$.

**Figure 3:** Linear closed–loop.

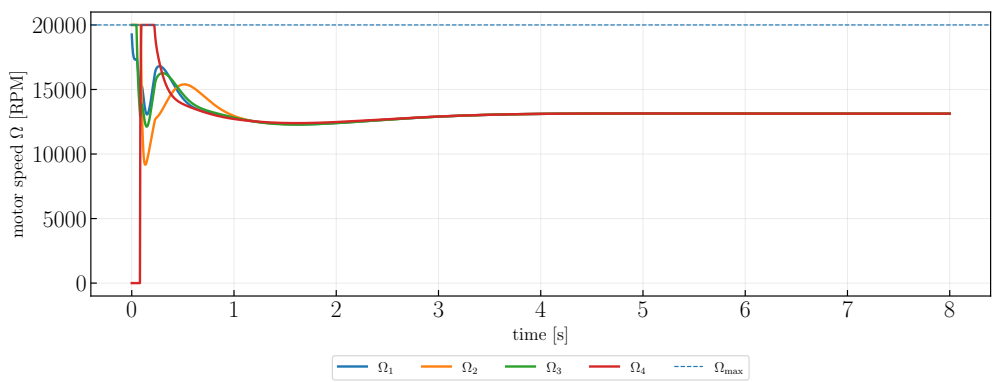
## 8.5 Linear MPC

We use $\Delta t_{\text{ctrl}} = 0.01$ s, horizon $N = 30$, weights $(Q, R)$ as in Eq. (8.7). The (ZOH) deviation model is

$$\delta x_{k+1} = A_d\, \delta x_k + B_d\, \delta u_k, \quad \delta x_0 = x_0 - x_s. \tag{8.10}$$




**(a)** States.




**(b)** Motor speeds $\Omega_i$.

**Figure 4:** Nonlinear closed–loop.

with

$$A_d = \begin{bmatrix} & & 0 & -4.9033\times10^{-4} & 0 & 0 & -1.6344\times10^{-6} & 0 \\ I_3 & 0.0100\, I_3 & 4.9033\times10^{-4} & 0 & 0 & 1.6344\times10^{-6} & 0 & 0 \\ & & 0 & 0 & 0 & 0 & 0 & 0 \\ & & 0 & -9.8067\times10^{-2} & 0 & 0 & -4.9033\times10^{-4} & 0 \\ \mathbf{0}_{3\times3} & I_3 & 9.8067\times10^{-2} & 0 & 0 & 4.9033\times10^{-4} & 0 & 0 \\ & & 0 & 0 & 0 & 0 & 0 & 0 \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & & I_3 & & & 0.0100\, I_3 & \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & & \mathbf{0}_{3\times3} & & & I_3 & \end{bmatrix}. \tag{8.11}$$

$$B_d = \begin{bmatrix} 0 & 0 & -1.2611\times10^{-7} & 0 \\ 0 & 1.8573\times10^{-7} & 0 & 0 \\ -3.3245\times10^{-5} & 0 & 0 & 0 \\ 0 & 0 & -5.0446\times10^{-5} & 0 \\ 0 & 7.4293\times10^{-5} & 0 & 0 \\ -6.6489\times10^{-3} & 0 & 0 & 0 \\ 0 & & & \\ 0 & \mathrm{diag}(2.2727\times10^{-3},\ 1.5432\times10^{-3},\ 9.1912\times10^{-4}) & & \\ 0 & & & \\ 0 & & & \\ 0 & \mathrm{diag}(4.5455\times10^{-1},\ 3.0864\times10^{-1},\ 1.8382\times10^{-1}) & & \\ 0 & & & \end{bmatrix}. \tag{8.12}$$

and the terminal $P_\infty$ from $\mathrm{DARE}(A_d, B_d, Q, R)$ was found to be:

$$P_\infty = \begin{bmatrix} 803.392 & 0.000 & 0.000 & 218.702 & 0.000 & 0.000 & 0.000 & -245.701 & 0.000 & 0.000 & -2.416 & 0.000 \\ 0.000 & 803.392 & 0.000 & 0.000 & 218.702 & 0.000 & 245.701 & 0.000 & 0.000 & 2.416 & 0.000 & 0.000 \\ 0.000 & 0.000 & 1227.181 & 0.000 & 0.000 & 521.353 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 218.702 & 0.000 & 0.000 & 150.555 & 0.000 & 0.000 & 0.000 & -193.716 & 0.000 & 0.000 & -1.927 & 0.000 \\ 0.000 & 218.702 & 0.000 & 0.000 & 150.555 & 0.000 & 193.716 & 0.000 & 0.000 & 1.927 & 0.000 & 0.000 \\ 0.000 & 0.000 & 521.353 & 0.000 & 0.000 & 531.555 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 245.701 & 0.000 & 0.000 & 193.716 & 0.000 & 511.950 & 0.000 & 0.000 & 5.159 & 0.000 & 0.000 \\ -245.701 & 0.000 & 0.000 & -193.716 & 0.000 & 0.000 & 0.000 & 511.950 & 0.000 & 0.000 & 5.159 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 103.572 & 0.000 & 0.000 & 1.300 \\ 0.000 & 2.416 & 0.000 & 0.000 & 1.927 & 0.000 & 5.159 & 0.000 & 0.000 & 0.839 & 0.000 & 0.000 \\ -2.416 & 0.000 & 0.000 & -1.927 & 0.000 & 0.000 & 0.000 & 5.159 & 0.000 & 0.000 & 0.839 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 1.300 & 0.000 & 0.000 & 0.917 \end{bmatrix} \tag{8.13}$$

and, $\lambda_{\min}(P_\infty) = 7.871\times10^{-1} > 0$, confirming positive definiteness.

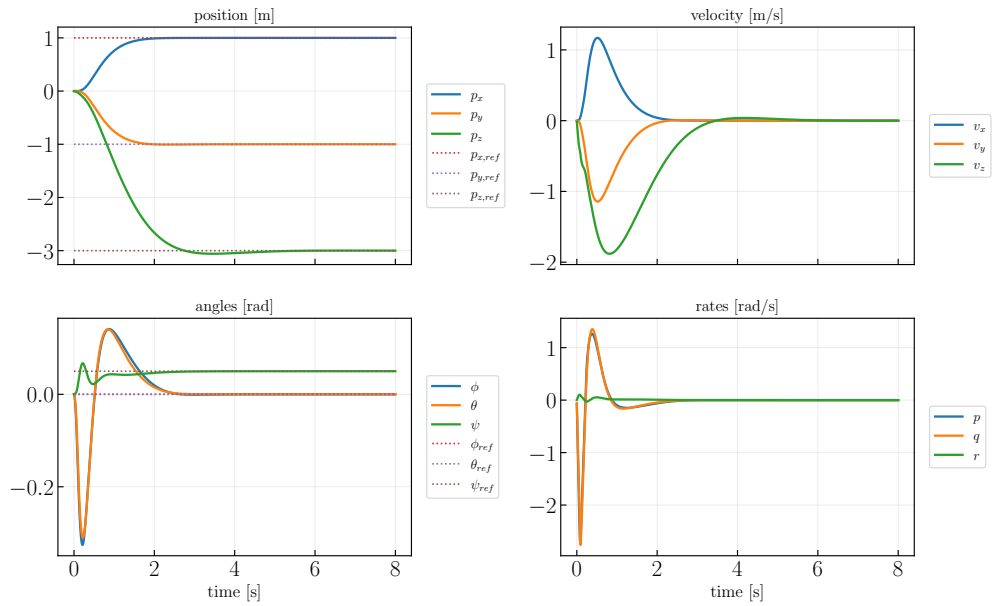
At each control step we solve a finite-horizon quadratic program on the ZOH deviation model $(A_d, B_d)$ with horizon $N = 30$ and sampling $\Delta t_{\mathrm{ctrl}} = 0.01$ s, using the same $(Q, R)$ as in Eq. (8.7) and a terminal cost with $P$ from $\mathrm{DARE}(A_d, B_d, Q, R)$. The optimizer returns the first deviation input which is applied to the nonlinear plant; RPM limits are enforced via the allocation/saturation map. The resulting state and motor-speed traces are shown in Fig. 5.

On the nonlinear plant the controller achieves $J = 66.72$ with input saturation on 110 of 4001 steps. Settling/overshoot (2% band): $p_x$ Ts = 1.824 s, OS = 0.00%; $p_y$ Ts = 1.678 s, OS = 0.61%; $p_z$ Ts = 2.522 s, OS = 1.94%; $\psi$ Ts = 2.662 s, OS = 34.76%. We report peak magnitudes: $|\phi|_{\mathrm{pk}} \approx 0.326$ rad, $|\theta|_{\mathrm{pk}} \approx 0.310$ rad.

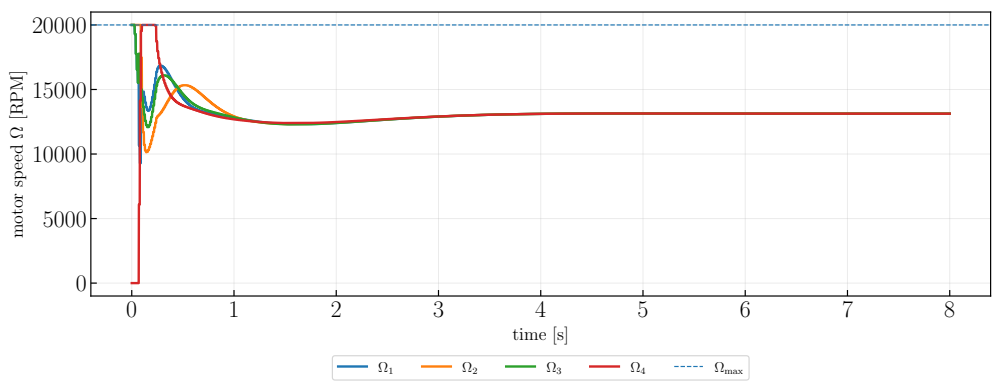
## 8.6 Nonlinear MPC

We use $\Delta t_{\mathrm{ctrl}} = 0.02$ s and horizon $N = 80$. The decision at each stage is $W_k = \left[\Omega_{1,k}^2, \Omega_{2,k}^2, \Omega_{3,k}^2, \Omega_{4,k}^2\right]^\top$ with bounds $0 \le W_{i,k} \le 4.0\times10^8$ (i.e., $0 \le \Omega_{i,k} \le 2.0\times10^4$ RPM), and the applied input is

$$u_k = \begin{bmatrix} T_k \\ \tau_{x,k} \\ \tau_{y,k} \\ \tau_{z,k} \end{bmatrix} = M\,W_k, \qquad M \text{ as in Eq. (8.2).}$$




**(a)** States.




**(b)** Motor speeds $\Omega_i$.

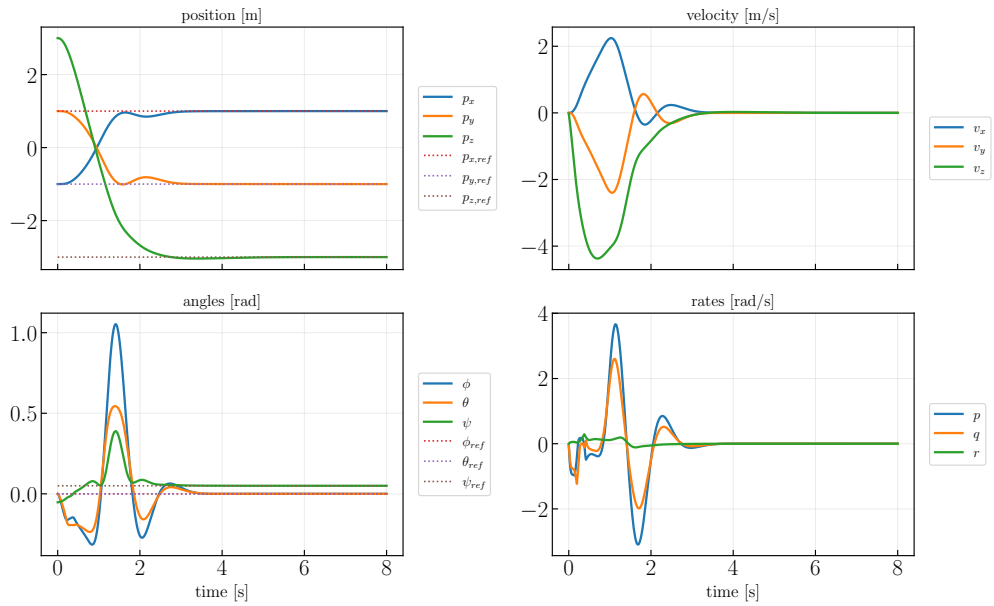
**Figure 5:** Linear MPC closed–loop.

Let $f(x, u)$ be the fully substituted continuous dynamics from Eq. (8.1). The one–step RK4 map used as the stage equality in the NLP is

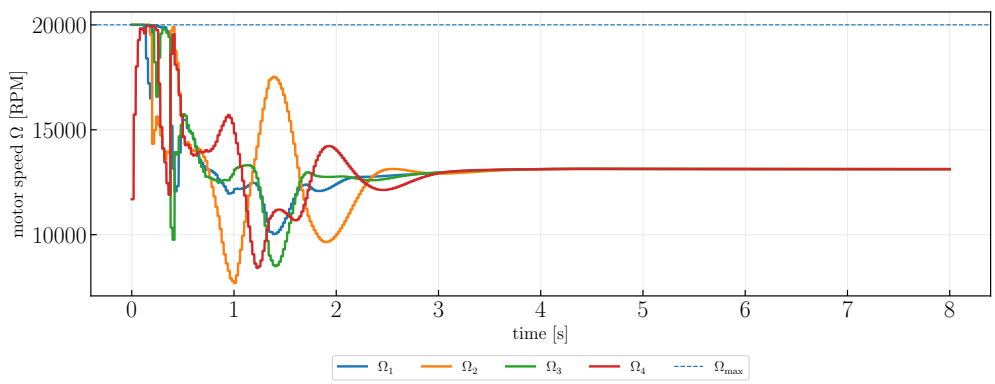
$$p_{k+1} = p_k + \frac{0.02}{6}\left(k_{1,p} + 2k_{2,p} + 2k_{3,p} + k_{4,p}\right), \tag{8.14a}$$
$$v_{k+1} = v_k + \frac{0.02}{6}\left(k_{1,v} + 2k_{2,v} + 2k_{3,v} + k_{4,v}\right), \tag{8.14b}$$
$$\eta_{k+1} = \eta_k + \frac{0.02}{6}\left(k_{1,\eta} + 2k_{2,\eta} + 2k_{3,\eta} + k_{4,\eta}\right), \tag{8.14c}$$
$$\omega_{k+1} = \omega_k + \frac{0.02}{6}\left(k_{1,\omega} + 2k_{2,\omega} + 2k_{3,\omega} + k_{4,\omega}\right). \tag{8.14d}$$

At each step the NMPC solves a nonlinear program over the horizon $N = 80$ with sampling $\Delta t_{\text{ctrl}} = 0.02$ s. The state update constraint uses the RK4 map in (8.14). We apply the first input and repeat (receding horizon). The closed–loop states and rotor speeds are shown in Fig. 6.

On the nonlinear plant, the nonlinear MPC gives $J = 255.85$ with input saturation on 110 of 4001 control




**(a)** States.




**(b)** Motor speeds $\Omega_i$.

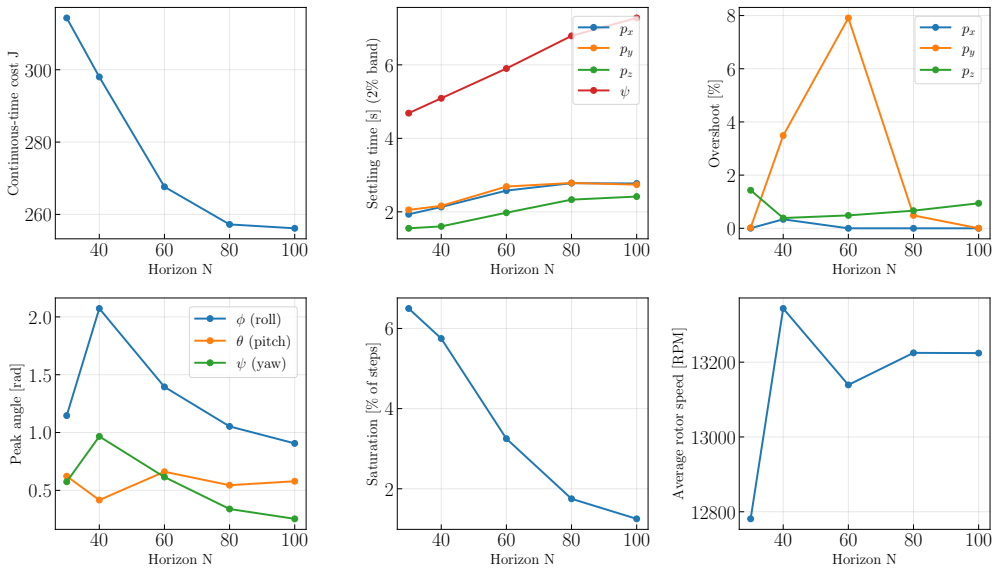
**Figure 6:** Nonlinear MPC closed–loop.

steps. Settling/overshoot (2% band): $p_x$ Ts = 2.716 s, OS = 0.00%; $p_y$ Ts = 2.690 s, OS = 0.00%; $p_z$ Ts = 2.404 s, OS = 1.61%; $\psi$ Ts = 2.646 s, OS = 164.32%. The yaw step is 0.05 rad, so 164% OS corresponds to a peak yaw error $\approx$ 0.13 rad. The peak values are: $|\phi|_{\text{pk}} \approx 0.728$ rad, $|\theta|_{\text{pk}} \approx 0.576$ rad. Performance could be improved by decreasing $\Delta t_{\text{ctrl}} = 0.02$ s and increasing the horizon $N$. However, the NMPC cost is high, and tighter sampling/horizon are subject to computational (time/memory) limitations.

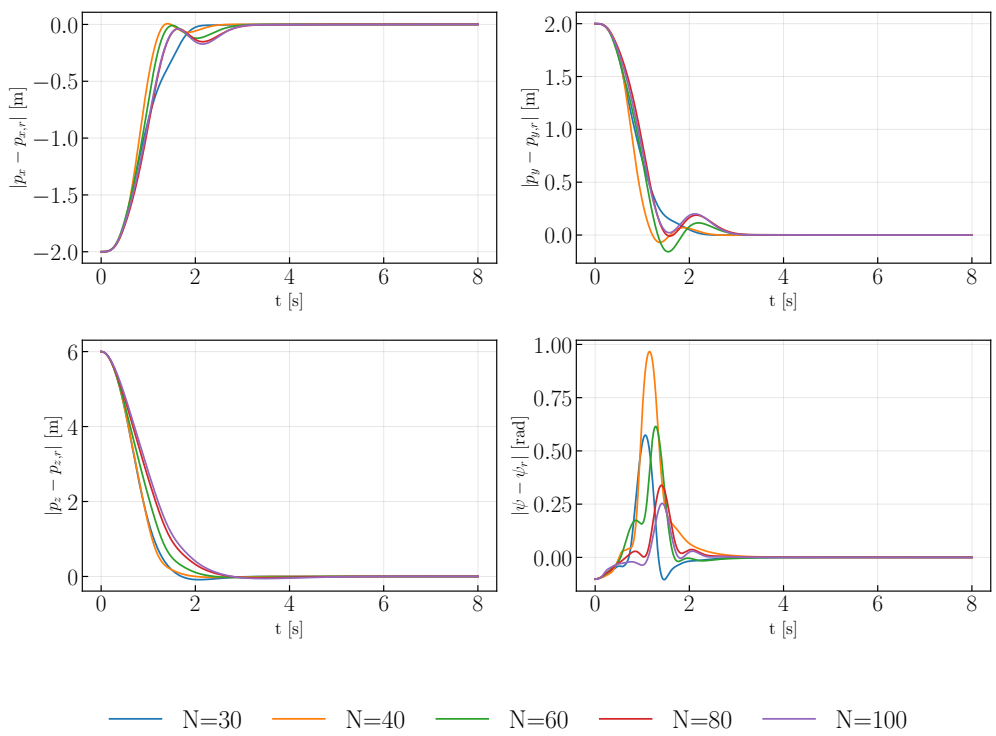
## 8.7 Sensitivity to Horizon and Actuator Limits

To assess robustness and to choose reasonable design points, we ran two sweeps on the nonlinear plant with the same controller structure, cost weights, sampling $\Delta t_{\text{ctrl}} = 0.02$ s, solver settings, and *fixed* initial condition as in the base case. The warm–start strategy (state/control repeats) was kept identical across runs. We varied (i) the prediction horizon $N \in \{30, 40, 60, 80, 100\}$, and (ii) the actuator bounds by scaling the maximum rotor speed $\Omega_{\max} \in \{14{,}000, \ldots, 19{,}000\}$ RPM while keeping $\Omega_{\min}$ unchanged. For each run we

report continuous–time cost $J$, 2%-band settling times, overshoot (for the tracked translational states), peak angles for $\phi, \theta, \psi$, the fraction of time steps with allocation saturation, and (for context) the average rotor speed. Summary charts and representative error time series are provided for both sweeps (see Figs. 7a, 7b, 8a and 8b).

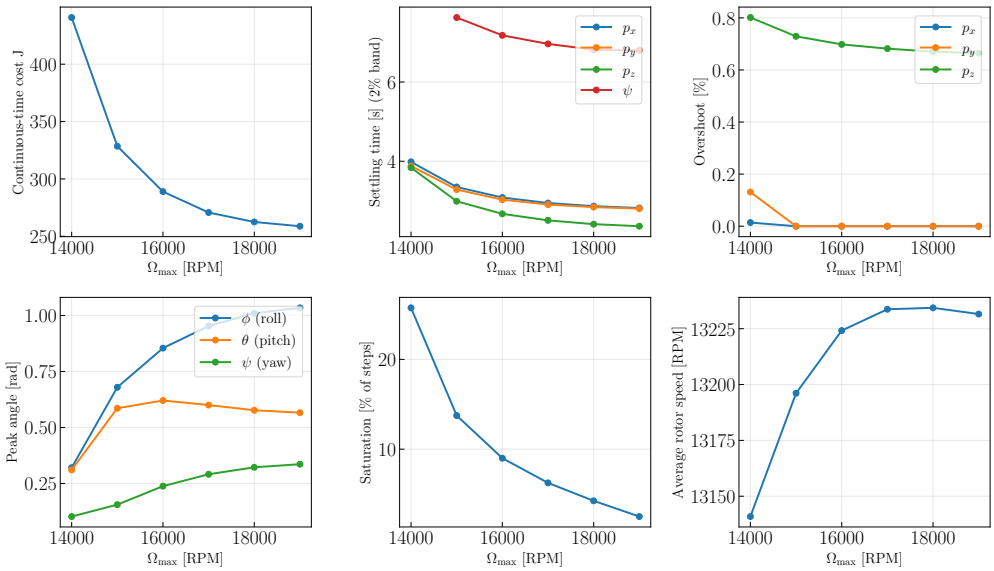



**(a)** Summary metrics vs horizon $N$ (cost, settling, overshoot, peak angles, saturation, avg. RPM).

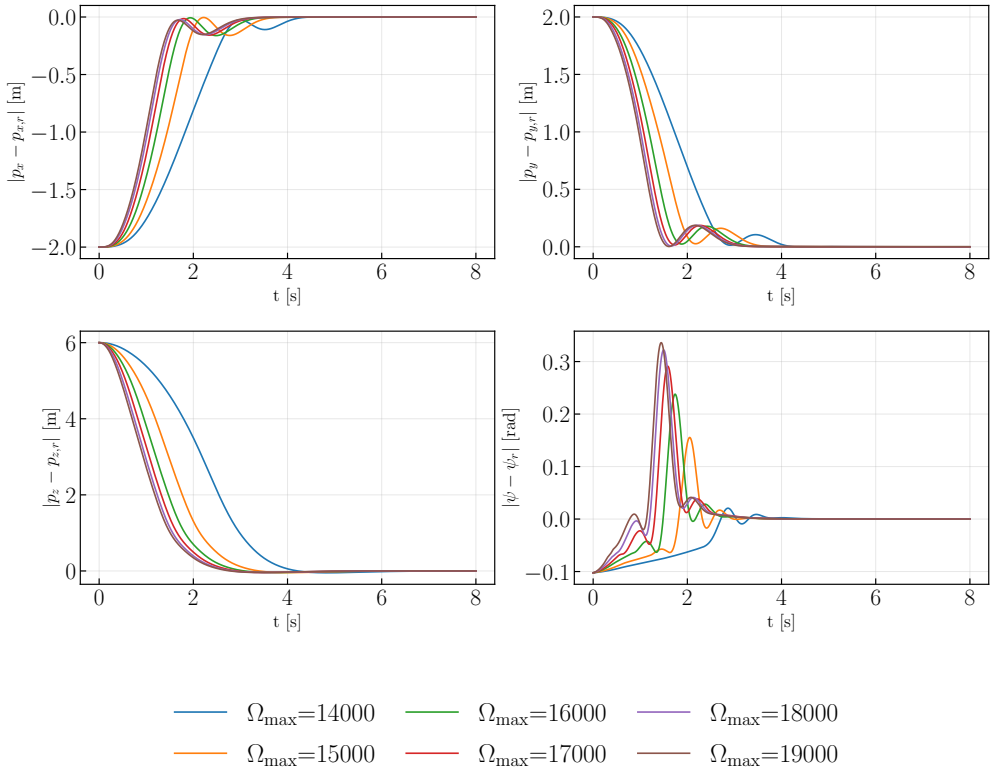



**(b)** Error overlays vs time for $p_x, p_y, p_z, \psi$ across $N$.

**Figure 7:** Nonlinear MPC horizon sweep. Panels summarize the sensitivity of cost and transients to the prediction horizon.




**(a)** Summary metrics vs actuator limit $\Omega_{max}$ (cost, settling, overshoot, peak angles, saturation, avg. RPM).




**(b)** Error overlays vs time for $p_x, p_y, p_z, \psi$ across $\Omega_{max}$.

**Figure 8:** Nonlinear MPC actuator–limit sweep. Panels summarize the sensitivity of performance to $\Omega_{max}$.

**Effect of the horizon $N$.** The cost $J$ decreases monotonically as $N$ grows (diminishing returns beyond $N \approx 80$), consistent with the added look–ahead improving setpoint capture (Fig. 7a, top–left). Settling times for $p_x, p_y, p_z$ remain clustered around 2–3 s, whereas $\psi$ settles more slowly and trends upward with $N$, reflecting a more cautious yaw transient under longer horizons (top–middle). Overshoot in translation is generally small,

with a localized spike on $p_y$ at an intermediate $N$, which matches the time–series overlays (top–right and Fig. 7b). Peak attitude angles drop as $N$ grows—especially $\phi$—indicating smoother attitude excursions with longer anticipation (bottom–left). The allocator saturates less often as $N$ increases (bottom–middle), and the average rotor speed rises modestly then levels off (bottom–right). Collectively, the curves suggest $N = 80$ as a good accuracy–vs–effort trade–off: low cost and saturation without materially longer transients.

**Effect of actuator limits $\Omega_{\max}$.** Relaxing the rotor speed bound reduces cost and generally shortens settling times for translation, while yaw shows milder improvement (Fig. 8a, top row). Translational overshoot remains near zero across the sweep (top–right). Interestingly, the *peak* attitude excursions $(\phi, \theta, \psi)$ grow with larger $\Omega_{\max}$ (bottom–left): with more available torque the controller corrects faster but briefly allows sharper attitude deflections before settling, a pattern also visible in the yaw error overlays (Fig. 8b). Saturation incidence drops steeply as $\Omega_{\max}$ increases (bottom–middle), confirming that tight bounds—not controller aggressiveness—were the source of early saturation. The average rotor speed changes little (bottom–right), indicating that gains in performance stem from increased *authority* rather than sustained higher RPM. These trends argue for selecting $\Omega_{\max}$ just high enough to avoid frequent saturation while keeping peaks acceptable for hardware safety.

**Time–domain view.** The error overlays for both sweeps show consistent behavior: $p_z$ is the fastest channel; $p_x, p_y$ exhibit small, well–damped bumps; and yaw transients peak around 1–2 s then decay smoothly (Figs. 7b and 8b). The spread across curves narrows as either $N$ increases or $\Omega_{\max}$ loosens, matching the metric trends.

**Design trade-off.** Within our compute budget and plant model fidelity, $N = 80$ with $\Omega_{\max}$ near the nominal 20,000 RPM bound yields a balanced design: low cost, minimal saturation, limited overshoot, and moderate peak attitudes. If yaw settling must be reduced further, options include (i) modest yaw weighting increases in $Q$, (ii) slightly higher $\Omega_{\max}$ where hardware allows, or (iii) a shorter $\Delta t_{\text{ctrl}}$ if real–time compute permits. The sweeps validate that performance is primarily limited by available actuation and solver budget, not by pathological closed–loop behavior.

## 8.8 Minimum Principle TPBVP

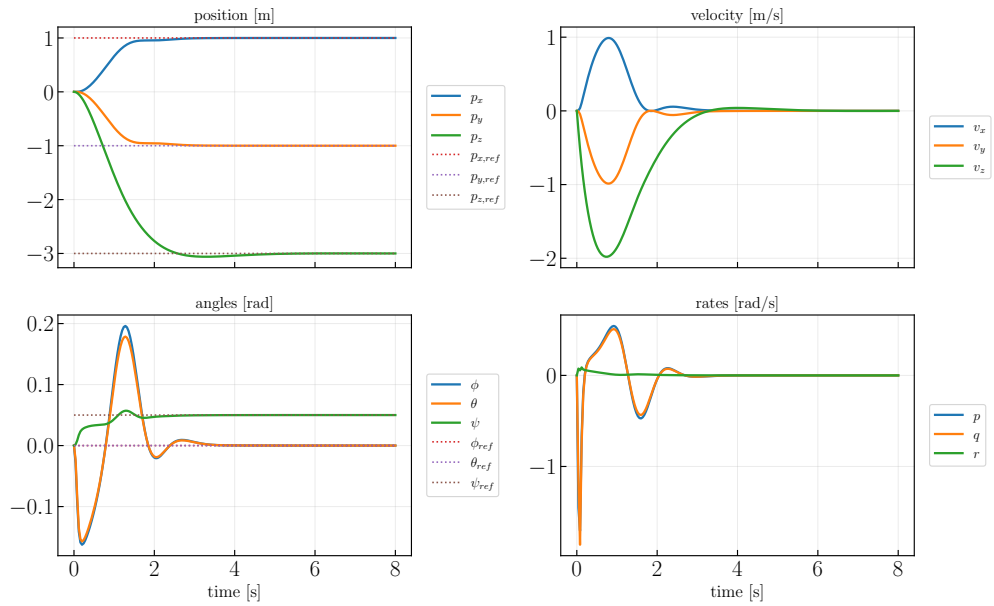
We solve a finite–horizon nonlinear OCP by *direct multiple shooting* (CasADi + Ipopt) on the full rigid–body model, using the same $(Q, R)$ and a constant reference. The shooting constraint uses the exact one–step RK4 map $\Phi$, so $x_{k+1} = \Phi(x_k, W_k^2, \Delta t)$. The objective is

$$\tfrac{1}{2}\sum_{k=0}^{N-1}\big((x_k - x_r)^\top Q (x_k - x_r) + (u_k - u_r)^\top R (u_k - u_r)\big)\Delta t \;+\; \tfrac{1}{2}(x_N - x_r)^\top Q_f (x_N - x_r),$$

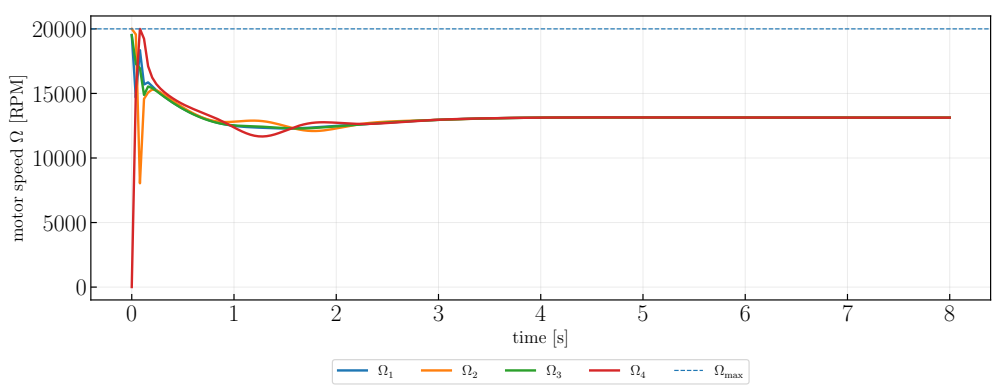
with $Q_f = 10\,Q$ and hard RPM bounds $0 \le \Omega_{i,k} \le 2.0{\times}10^4$ RPM. For diagnostics we form exact Jacobians of $\Phi$ to check discrete PMP conditions. The state and motor–speed trajectories are in Fig. 9, and the costates/residuals are in Fig. 10.

The computed solution attains $J = 62.79$ with no active RPM limits (0/201 steps). Step responses (2% band): $p_x$ and $p_y$ both settle at Ts = 2.600 s with OS = 0.00%; $p_z$ settles at Ts = 3.440 s with OS = 2.04%; $\phi$ and

$\theta$ are conservative with Ts = 2.120, 1.800 s and OS = 0.00%. Peak (wrapped) angles are $|\phi|_{\text{pk}} \approx 0.196$ rad, $|\theta|_{\text{pk}} \approx 0.179$ rad, and $|\psi - \psi_r|_{\text{pk}} \approx 0.050$ rad.




**(a)** States.




**(b)** Motor speeds $\Omega_i$.

**Figure 9:** Nonlinear OCP (PMP/Ipopt) closed–loop.

Alongside the optimal states and motor speeds, we report costates and first–order optimality residuals (Fig. 10a). Let $\Phi(x_k, \Omega_k^2, \Delta t)$ be the exact one–step map used in multiple shooting with Jacobians $F_x := \partial\Phi/\partial x$ and $F_{\Omega^2} := \partial\Phi/\partial\Omega^2$. With $u_k = M\Omega_k^2$ and reference $(x_r, u_r)$, the discrete PMP checks are:

$$r_{u,k} := M^\top R\,(u_k - u_r)\,\Delta t \;+\; F_{\Omega^2}(x_k, w_k^2)^\top \lambda_{k+1}, \tag{8.15}$$

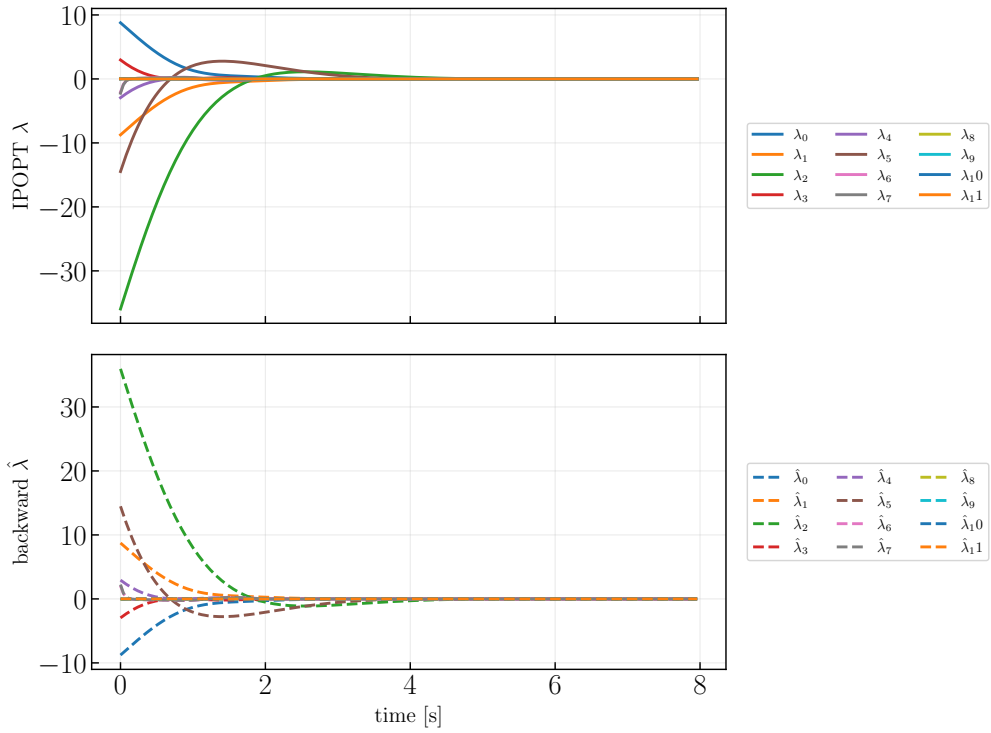
$$\hat{\lambda}_k := F_x(x_k, \Omega_k^2)^\top \hat{\lambda}_{k+1} + Q\,(x_k - x_r)\,\Delta t, \quad \hat{\lambda}_N := Q_f\,(x_N - x_r), \tag{8.16}$$

$$r_{p,k} := \lambda_{k+1}^{(\text{IPOPT})} - \hat{\lambda}_{k+1}. \tag{8.17}$$

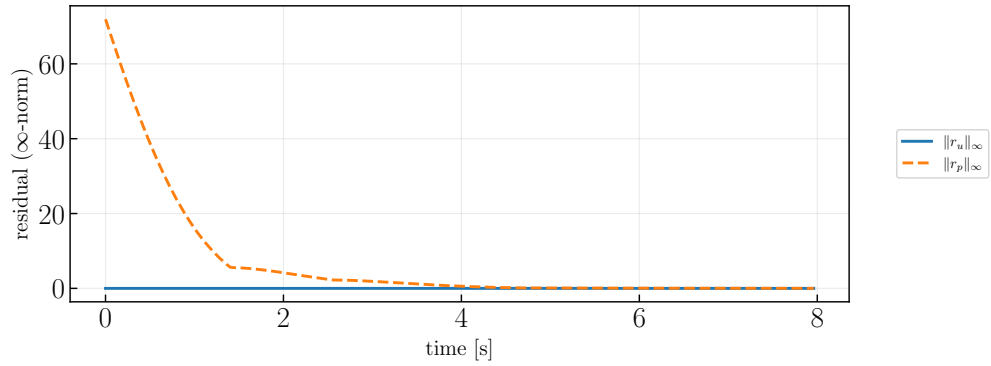
Here $r_u$ enforces stationarity w.r.t. the decision variable $\Omega^2$ (squared rotor speeds), while $r_p$ compares IPOPT multipliers on dynamics defects to the backward–recursed costates.

Stationarity is satisfied to numerical precision: $\max_k \|r_u\|_\infty = 3.97\times10^{-10}$. The multiplier consistency

residual $\|r_p\|_\infty$ is largest near the beginning of the horizon and decays rapidly, consistent with using IPOPT multipliers (from the defect constraints) versus a separate backward recursion for $\hat{\lambda}$; the terminal mismatch is small ($1.28\times10^{-2}$). No hard RPM bounds were active (0/201), so the improvement in $J$ versus LQR comes from horizon-wide effort allocation rather than constraint handling.




**(a)** Costates: IPOPT multipliers $\lambda$ (top) and backward recursion $\hat{\lambda}$ (bottom).




**(b)** Optimality residuals over the horizon: $\|r_u\|_\infty$ (stationarity) and $\|r_p\|_\infty$ (multiplier consistency).

**Figure 10:** PMP diagnostics for the nonlinear OCP.

## 8.9 Summary and Comparison

Table 2 consolidates the closed–loop metrics for all controllers using the same $(Q, R)$ and step reference. The direct multiple–shooting OCP (PMP/Ipopt) attains the lowest cost while keeping translations deliberately smooth and with zero input saturation. Infinite–horizon LQR is competitive on cost but does activate bounds

briefly (sat. steps reported). Linear MPC lands close to LQR in cost, while NMPC yields the largest yaw overshoot. The coarse–grid nonlinear MPC, with the chosen sampling/horizon, is markedly more aggressive in attitude (large peak angles) and much worse in cost; this is consistent with the tighter grid/horizon requirements of NMPC on fast full dynamics.

**Table 2:** Unified comparison: objective $J$, input saturation count, 2%–band settling/overshoot for translations and yaw, and peak roll/pitch.

| Controller / Plant | $J$ | Sat. steps | $p_x$ | | $p_y$ | | $p_z$ | | $\psi$ | | Peaks [rad] | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | $T_s$ [s] | OS [%] | $T_s$ [s] | OS [%] | $T_s$ [s] | OS [%] | $T_s$ [s] | OS [%] | $\lvert\phi\rvert_{\text{pk}}$ | $\lvert\theta\rvert_{\text{pk}}$ |
| LQR (linear plant) | 63.62 | 89 | 2.004 | 0.00 | 2.008 | 0.00 | 3.600 | 2.19 | 1.976 | 0.00 | 0.319 | 0.319 |
| LQR (nonlinear plant) | 67.15 | 109 | 1.826 | 0.00 | 1.696 | 0.56 | 2.528 | 1.95 | 2.706 | 16.40 | 0.319 | 0.307 |
| Linear MPC (nonlinear) | 66.72 | 110 | 1.824 | 0.00 | 1.678 | 0.61 | 2.522 | 1.94 | 2.662 | 34.76 | 0.326 | 0.310 |
| Nonlinear MPC (nonlinear) | 255.85 | 110 | 2.716 | 0.00 | 2.690 | 0.00 | 2.404 | 1.61 | 2.646 | 164.32 | 0.728 | 0.576 |
| OCP (PMP/Ipopt, nonlinear) | 62.79 | 0 | 2.600 | 0.00 | 2.600 | 0.00 | 3.440 | 2.04 | 2.560 | 14.18 | 0.196 | 0.179 |

# 9 Conclusion

This report compared four optimal controllers near hover: infinite-horizon LQR; linear MPC on a ZOH discretization with a DARE terminal weight; NMPC on the full model with RK4 multiple shooting; and a nonlinear OCP via PMP/Ipopt. All used the same 12-state model, identical allocation/limits, and a shared quadratic cost.

On the nonlinear plant, PMP achieved the lowest objective with smooth translations and no actuator saturation—an upper bound when computation permits. LQR and linear MPC showed similar metrics and competitive cost with brief, modest saturation, making them practical for real-time use. With the coarse grid, NMPC had higher cost and larger attitude/yaw peaks, indicating a need for finer sampling or longer horizons.

Sensitivity indicates that a horizon $N \approx 80$ and bounds $\Omega_{\max} \approx 2 \times 10^4$ RPM balance accuracy and effort. Yaw transients improve with slightly higher yaw weight, relaxed $\Omega_{\max}$ where hardware allows, or a smaller control step $\Delta t_{\text{ctrl}}$. Overall, LQR/LMPC are strong real-time baselines; PMP provides the best performance; NMPC benefits markedly from denser grids.